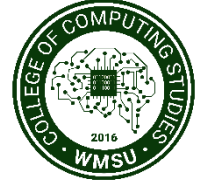




Republic of the Philippines
Western Mindanao State University
College of Computing Studies
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Zamboanga City



HEAREASE: OPTIMIZING BARANGAY HEARING SCHEDULING AND PREDICTING CASE RESOLUTION TIME USING DEEP NEURAL NETWORK

A Thesis Presented to the Faculty of
Department of Computer Science
College of Computing Studies

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Computer Science

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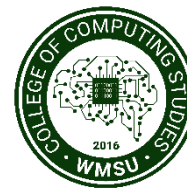
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May 2026



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Approval Sheet

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Abstract

Barangay hearing scheduling has to be more effective due to the growing complexity of legal cases in Philippine communities. This thesis presents the HearEase system, which uses deep neural networks to optimize the scheduling of barangay hearings and predict case resolution times. The study recognizes a number of flaws in traditional scheduling techniques that frequently lead to both unresolved cases and postponed hearings. HearEase, a pilot project in Barangay Tetuan, Zamboanga City, Philippines, was chosen because of the community's active involvement, the volume of cases it handles each year, and their willingness to fully cooperate with the study.

By utilizing various advanced algorithms, HearEase aims to provide barangay staff with a user-friendly interface which automates scheduling per historical data with predictive analytics. The methodology involves gathering data from past hearings, case results, and the implementation of deep learning models for prediction of future case resolutions. Initial outcomes indicate that the implementation of HearEase has the capacity to enhance the effectiveness of barangay hearings, ensuring resolutions of cases in a timely manner. In addition to reducing the administrative load on the Lupong Tagapamayapa and barangay employees, this system seeks to enhance the accessibility of justice for the local populace.

Keywords: Philippine governance, barangay hearings, deep neural network, resolution prediction, scheduling optimization

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CHAPTER I

INTRODUCTION

Background of the Study

With its function as the first level of government, barangay hearings play a vital role in resolving conflicts and maintaining community peace in the Philippines. Though important, long wait times and scheduling issues often call into question the effectiveness of barangay hearings. Especially for people in communities who depend on quick and fair choices, these elements disturb the tranquility of the community and act as obstacles to justice. Ignoring these obstacles could make barangay hearings face challenges in meeting community needs and offering prompt case resolutions, therefore hindering social progress.

Often run traditionally by writing everything down, which is error-prone and time-consuming, barangay hearings face one of the biggest obstacles to success. Resolving a case presented to the barangay hearing format on an ad hoc basis creates more problems and drags out the dispute since the mode of justice becomes ineffective. The unwieldy nature of justice diminishes confidence in the entire system because residents do not see the outcome of a hearing in a timely manner.

On top of this, there are no new solutions that can improve scheduling, the timing of resolutions, and resource allocation. Automated systems would help

reduce human error and inefficiencies in barangay hearings, which can lead to delays and unresolved cases. The inability to address this variety of challenging situations could be inconvenient for residents and barangay officials, especially when they feel overwhelmed by several cases that continue to grow in volume with limited resources. The lack of technological solutions means that there will not be enough improvements made for the efficiency of hearings and that delays and frustrations continue in the community.

Many studies have looked at how technology can be used in ways to increase efficiency in administrative processes in the public sector. For example, a study evaluated the effective ways to implement AI-based systems for the municipality of Carmona, Cavite. The incorporation of automation increased the number of resource allocations, thus minimizing delays due to inefficient manual processes [1]. Another study conducted by Balilia et al. looked at how implementing automated systems in barangay offices improved administrative function related to automating project monitoring and record keeping as well. Notable improvements in efficiency increased public service delivery to residents [2]. The authors propose a more comprehensive system that focuses on administrative functions such as monitoring projects and processing documents, while excluding the consideration of barangay hearing procedures from the study's scope.

Melnik and Nasonov studied how Deep Neural Networks (DNN) and Reinforcement Learning can be used for scheduling optimization in general

workflow systems. They did report slightly positive results in which workflow assignments were enhanced and reduced processing times [3]. While they provide evidence that could demonstrate a positive improvement to several workflows developed, it has not been tested in other administrative environments, specifically barangay hearings, which involve a high structure of procedures, complex logics, and limits in resources.

Alternatively, a relevant study examined the use of Artificial Intelligence (AI) as a supplementary tool in judicial hearings. Their study goes further by proposing that AI could also assist in legal practice, particularly in adjudicating disputes. They suggested that AI possess the possibility to contribute to legal expertise or judgement, and even envisioned scenarios where cases could be fully resolved by a robotic judge handling the entire adjudication process. However, the issue of fairness was mostly limited to the extreme position of AI providing the full resolution outcome. They suggested that working with artificial intelligence as a support for decision-making would be fitting to apply human expertise [4].

Lastly, Bringula et al. examined the creation of e-Barangay, a web-based system meant to improve local government at the Philippines' barangay level. Their research developed an e-Barangay framework with elements including complaint processing, document requests, reminders and alerts. Using both subjective and objective criteria, the system's usability was assessed. The results revealed that e-Barangay satisfied local needs. The e-Barangay system

efficiently completed tasks in objective measurements and had positive feedback in subjective evaluations [5].

This study presents the development of HearEase, a web-based platform that leverages a Deep Neural Network (DNN) to optimize the scheduling of barangay hearings and predict case resolution timelines.

Designed for web accessibility, the system features distinct, role-based portals for barangay officials and a transparent mechanism for citizens to file complaints and track case status. By automating the scheduling process according to established Katarungang Pambarangay rules, the system aims to enhance administrative efficiency, minimize procedural delays, and facilitate the delivery of timely justice to the community.

This system will begin implementation in Barangay Tetuan in Zamboanga City for several important reasons. To begin with, Barangay Tetuan is among the largest barangays in Zamboanga City in terms of caseload and will therefore provide an opportunity to test the impact of the program without being vastly limited by its size. Having also a diverse population means that we can expect a broad selection of cases that can be used to assess the impact of the program across different demographics. Additionally, the local officials in Barangay Tetuan expressed that they will also provide a receptive environment for technological improvements in administrative processes. Their local officials welcomed the idea of incorporating technology to strengthen administrative processes in their

Barangay. Lastly, Barangay Tetuan produces good records of case hearings and resolutions. Therefore, available data points to train the AI system in the most effective way will be accessible. The study wishes to use Barangay Tetuan as a pilot study to collect effective feedback and outputs for disseminating the study to other barangays in the same region in the future.

Statement of the Problem

While the barangay justice system plays a vital role in local dispute resolution, Barangay Tetuan continues to rely on traditional, manual scheduling processes that are often unstructured and susceptible to human error. This reliance on ad-hoc methods frequently results in scheduling conflicts, resource mismanagement, and inconsistent estimation of hearing durations.

Consequently, these inefficiencies lead to procedural delays and potential inequity in accessing swift justice for complainants. Currently, there is a lack of data-driven mechanisms that utilize historical patterns to optimize hearing schedules and predict case resolution timelines, leaving the barangay unable to cope effectively with its increasing caseload.

Objectives

This study aims to build an artificial intelligence-driven system employing Deep Neural Networks (DNN) to maximize barangay hearing schedules and forecast case resolution times.

Specifically, the study will:

- Analyze the existing case resolution workflows and scheduling constraints in Barangay Tetuan to identify requirements for automation.
- Collect case data to be used to train the neural network for schedule prediction.
- Develop HearEase, a web-based information system that integrates AI-driven scheduling and case management features.
- Evaluate the predictive accuracy of the DNN model in forecasting hearing frequency and resolution timelines.
- Assess the system's usability, efficiency, and user satisfaction with the Lupong Tagapamayapa compared to traditional manual procedures.

Scope and Limitations

Scope of the Study

This study focuses on the design, development, and deployment of "HearEase," a web-based hearing scheduling and management system integrated with a Deep Neural Network (DNN) model. The primary objective is to

optimize the administrative workflow of the Katarungang Pambarangay by automating scheduling procedures and predicting case resolution timelines based on historical data patterns.

The study is conducted within the specific context of Barangay Tetuan, Zamboanga City. The system is designed to serve two distinct user groups:

1. Barangay Administrators (Punong Barangay, Barangay Secretary, and Lupong Tagapamayapa Secretary)

The system provides them with a dashboard for digital case filing, automated hearing scheduling, notification management, and analytical reporting.

2. Constituents (Residents)

The system provides a public-facing portal for filing complaints online and tracking the real-time status of their cases.

Technically, the system's scope covers the full cycle of case management—from the filing of the complaint to the issuance of the final certification (Settled, Arbitration, or Certificate to File Action). The predictive capabilities are limited to estimating the number of hearings and resolution duration using a dataset constructed from Barangay Tetuan's case records.

Limitations of the Study

To ensure the feasibility and focus of the research, the following limitations are established:

1. Geographical and Contextual Limit

The system is customized specifically for the operational workflow and volume of Barangay Tetuan. While the underlying logic may be applicable elsewhere, the dataset used to train the AI is localized; therefore, the prediction accuracy may not be immediately generalizable to other barangays with different demographic profiles without re-training.

2. Administrative vs. Judicial Role

The AI model is strictly limited to administrative scheduling and time estimation. It does not attempt to predict the legal outcome of a case nor suggest specific settlement terms. The discretionary power to mediate and decide on cases remains exclusively with the Lupon Tagapamayapa, in accordance with the Local Government Code.

3. Jurisdictional Limit on Fines

The system's case management is limited to disputes falling within the Katarungang Pambarangay's jurisdiction, specifically those that do not

involve a fine exceeding Five Thousand Pesos (₱5,000), as per the rules of the Local Government Code.

3. Parties in Conflict

The system will only process complaints between natural persons residing or working in Barangay Tetuan. It will not cater to complaints filed by or against organizations, corporations, or non-residents of the barangay. This limitation is necessary because the jurisdiction of the Katarungang Pambarangay, which HearEase is designed to support, is generally confined to disputes between natural persons who are residents of the same, contiguous, or specific barangays, and does not extend to juridical entities like corporations, pursuant to the Local Government Code of 1991.

4. Procedural Adherence

While the system allows for online filing, it does not replace the legal requirement for face-to-face confrontation. The system serves as a logistical tool to facilitate the scheduling of these in-person meetings. The issuance of summons and certifications will still be physically printed and served by Barangay Tanods as required by law, though the documents are generated digitally.

5. User Adoption

The system is web-based and requires an internet connection to function. The study evaluates the usability and efficiency of the system but does not measure the long-term behavioral change of the residents regarding digital adoption. The effectiveness of the system relies on the willingness of the barangay staff to transition from manual logbooks to digital encoding.

Significance of the Study

The development and implementation of HearEase offers significant contributions to the modernization of the barangay justice system. By integrating Artificial Intelligence into local governance, this study yields specific benefits for the following stakeholders:

For the Barangay Officials and Lupong Tagapamayapa

As the primary users, this system directly addresses their administrative burden. HearEase streamlines the hearing scheduling process by minimizing conflicts and human errors, providing reliable administrative support. By automating complex logistical tasks, officials can optimize their time management and focus their energy on substantive dispute resolution rather than clerical challenges. This shift from manual to digital operations is anticipated to improve workflow efficiency and overall job satisfaction.

For the Community and Local Residents

The system promotes a more accessible and transparent avenue for justice. By ensuring hearings are scheduled efficiently and communicated clearly, residents are spared the inconvenience of long wait times or unannounced cancellations. A more organized justice system fosters public trust, encouraging citizens to utilize proper legal channels for dispute resolution rather than resorting to informal or escalating conflicts.

For the Local Government Units (LGUs) and DILG

This study serves as a proof-of-concept for digital transformation at the grassroots level. The data generated by HearEase provides LGUs with valuable insights into crime trends, case volumes, and resolution rates, enabling data-driven decision-making and policy formulation. Furthermore, this system demonstrates how technology can enhance transparency and accountability in public service delivery.

For Future Researchers and the Academic Community

This research contributes to the growing body of knowledge regarding GovTech (Government Technology) and the application of Deep Neural Networks (DNN) in public administration. It serves as a foundational reference for future studies aiming to digitize other barangay

processes. The methodologies and datasets established in this study provide a benchmark for researchers attempting to solve similar scheduling and resource allocation problems in the Philippines.

Definition of Terms

Table 1: Definition of Terms

Term	Definition
1. Barangay	The smallest political unit in the Philippines, performing the grassroots level of governance responsible for local neighborhoods, including public hearings and local officials.
2. Conciliation Hearing	Less formal than arbitration, this is a dispute resolution process where Lupong Tagapamayapa facilitate discussions between disputing parties in order to assist them in arriving at a voluntary settlement. In contrast to mediation, the barangay officials can also provide possible solutions for the parties but leave the decision for them to settle the compromise agreement.
3. Conversational User Interfaces (CUIs)	Allow users to interact with computers or systems using natural language, either through text or voice, similar to chatting with a person. Examples

Term	Definition
	include chatbots and voice assistants like Siri or Alexa.
4. Deep Neural Network (DNN)	A kind of artificial intelligence that emulates the way the human brain learns. It processes information through several layers, enabling it to detect patterns and make predictions. It can make realistic approximations, meaning it can learn and model complex relationships between input and output.
5. Department of Trade and Industry (DTI)	A government agency in the Philippines responsible for spearheading trade, industry, and investment. It helps businesses grow, protects consumers, and supports economic development through policies, programs, and services.
6. Explainable AI (XAI)	An AI that can clearly explain how it makes decisions. Instead of just giving an answer, it shows the reasons behind it, making it easier for people to understand and trust.

Term	Definition
7. Institutional Service Providers (ISPs)	Organizations that provide important services such as legal, financial, and tech support for businesses, communities, and even governments. They can be local, national, or even global. These service providers play a significant part in driving economic growth, putting policies into action, and making sure things run smoothly across different sectors.
8. Local Government Unit (LGU)	LGU manages specific areas, like provinces, cities, or even smaller communities like barangays in the Philippines. The Local Government Units (LGUs) operate under the Department of the Interior and Local Government (DILG). They are vested with a fair amount of independence when it comes to handling local matters. This includes, public services, law enforcement, infrastructure, health, and education.

Term	Definition
9. Lupon Tagapamayapa	A local conflict resolution group in the Philippines under the Barangay Justice System. This unit is made up of barangay officials along with a few community members they appoint. Their main job is to mediate and sort out disputes right at the barangay level, which helps keep those minor issues from clogging up the courts.
10. Mediation Hearing	Informal sessions that are spearheaded by the Lupon Tagapamayapa. Mediation Hearing is the three (3) initial hearings where the Lurons try to reconcile the complainant and the respondent to avoid further fluctuation of the case.
11. Natural Language Processing (NLP)	A branch of Artificial intelligence that helps computers to understand, interpret, and even create sentences like a human. NLPs are commonly used for chatbots, translation services, and voice assistants.

Term	Definition
12. Reinforcement Learning (RL)	A type of AI that learns through experience through trial and error. Computers try things out, get feedback, and adjusts their actions to get better results.
13. Scheduling Algorithm	A set of guidelines and rules that automated systems use to figure out how to best allocate time slots and resources for different events and appointments.
14. Technology-Organization-Environment (TOE) Framework	The TOE Framework looks at three big factors: first, there technology itself, second, the organization, which includes how things are structured and the resources available and lastly, the environment, which pertains to market trends and regulations. These pieces come together to help organizations figure out the best way to assess the factors that influence the adoption of a new technology.

Term	Definition
15. User Interface (UI)	This is how a user interacts with a system. It scopes everything from visual elements like buttons, screens, and navigation tools that are designed to make User Experience (UX) as smooth as possible.

CHAPTER II

REVIEW OF RELATED LITERATURE

Related Studies

The Barangay Justice System, or *Katarungang Pambarangay*, serves as the Philippines' oldest and most community-centered dispute resolution mechanism. Established by Presidential Decree No. 1508 in 1978 and later strengthened through Republic Act 7160 (Local Government Code of 1991), it institutionalized the practice of conciliating minor disputes at the grassroots level before elevating them to higher courts [6]. Designed to ease judicial backlogs and promote harmony among neighbors, the system underscores mediation, conciliation, and arbitration as primary tools of justice .

Contemporary research regards the BJS as one of the most adaptive community-based justice systems in Southeast Asia. Medina and Agustin describe it as a hybrid legal institution, merging traditional consensus-building with statutory frameworks that empower the *Lupon Tagapamayapa*—a council of mediators appointed by the barangay captain—to pursue amicable settlements [7]. A 2025 policy study from the Philippine Information Agency reported the rollout of new Barangay Justice System modules across Mindanao, aiming to integrate legal frameworks with cultural practices and strengthen peace and governance at the barangay level [8].

The Ministry of Interior and Local Government in the Bangsamoro Autonomous Region in Muslim Mindanao (MILG-BARMM) has invested resources in modernizing local governance, training barangay officials in mediation, and creating digital knowledge products for dispute monitoring. Despite these advancements, inefficiencies persist in manual documentation, record retrieval, and hearing scheduling—problems now being addressed through emerging AI-driven innovations such as Barangay Automated Records Solution (BARS), which digitizes dispute management and enhances transparency [9].

This review contextualizes the *HearEase* project within recent developments by organizing the literature under key domains: Artificial Intelligence-Driven Systems, Deep Neural Networks, Case Resolutions and Barangay Hearings, and Conventional Approaches. Each subtopic situates previous work within theoretical, technical, and contextual frameworks relevant to barangay-level governance.

Artificial Intelligence-Driven Systems

Scholarly attention toward artificial intelligence in governance has accelerated in the past decade, emphasizing automation's role in policy decisions and administrative workflows. Studies by Hilbert et al. highlight how AI governance enhances process accuracy, enabling predictive and data-informed decision-making within public institutions [10]. Similarly, Rowe identifies that

integrating ethical frameworks into AI supervision fosters transparency and accountability [11].

In the Philippines, the Department of Trade and Industry's *National AI Strategy Roadmap* promotes the use of machine learning for administrative optimization, encouraging LGUs to adopt AI-assisted document and case systems. In Zamboanga Peninsula, Zabala reported that AI deployments in LGUs boosted citizen engagement and data accuracy, particularly in complaint tracking [12]. These findings mirror innovations such as the SLSU-LGU-PNP AI-Driven Municipal Crime System (2025), which leveraged predictive analytics for incident scheduling and management, proving the feasibility of AI in localized governmental systems [13].

However, Rjab et al. caution that successful AI adoption requires addressing technical, environmental, and organizational barriers, including staff resistance and limited ICT literacy [14]. This aligns with the *HearEase* project's aim of designing an adaptive, user-friendly system that empowers barangay officials while retaining procedural transparency.

Deep Neural Networks

Deep Neural Networks (DNNs) represent one of the most advanced forms of artificial intelligence used in predictive analytics and scheduling optimization. Melnik and Nasonov showcased how coupling DNN models with Reinforcement Learning substantially improves dynamic workflows [15]. Building on such

concepts, Bouška, Šůcha, and Hanzálek presented a deep learning-driven scheduling algorithm that minimized tardiness in industrial timetables to near-optimal outcomes. Their research demonstrated that DNNs can autonomously learn scheduling priorities from data—an idea directly transferable to barangay hearing scheduling [16].

Locally, Belacho et al. at National University Dasmariñas developed a Barangay Information Management System (BIMS) applying early machine-learning elements to automate data logging. This represents the preliminary phase of integrating DNNs for Philippine governance systems. The iterative learning and adaptive optimization techniques within *HearEase* extend these principles to case resolution prediction, combining historical barangay hearing data with localized socio-legal features to forecast case durations more accurately. [17].

Case Resolutions and Barangay Hearings

The Katarungang Pambarangay ensures accessible, low-cost alternatives to formal litigation, but inefficiencies in manual processes still impede timely justice delivery. Respicio outlines that barangay complaint proceedings encompass scheduling summonses, mediation, conciliation, and arbitration—steps vulnerable to delay without digital tracking [18]. The International Research Journal of Science, Technology, Education, and

Management found that community satisfaction hinges heavily on timely hearings, yet limited digitization slows outcome recording .

Medina and Agustin observed that integration of cultural traditions with statutory law makes barangay justice unique but also fragmented when inconsistently applied [19]. BARMM's 2025 knowledge modules and the Bangsamoro Rollout Program have demonstrated how integrating legal guidance with local governance training revitalizes community-based justice [20]. Together with DNN predictive modeling, these developments affirm how *HearEase* can modernize BJS by ensuring accurate scheduling, equitable access, and consistent resolution tracking across barangays.

Conventional Approaches

Most barangays continue to rely on conventional manual systems—bound ledgers, handwritten minutes, and physical summons logs—that often lead to lost records and procedural redundancies. Porcincula documented these issues among Laguna municipalities, noting frequent duplication of entries and missing documentation [21]. The Barangay Automated Records Solution (BARS) initiative responded by digitizing hearings and document archives, improving law-enforcement coordination and data transparency [22].

Furthermore, mainstream jurisprudence studies from the Philippine Law Journal revisit BJS legal frameworks, recommending *electronic case lodgement* to lessen judiciary case backlogs . Without automation, barangay officials face

administrative overload, inconsistent scheduling, and limited visibility of case histories. Hence, transitioning toward AI-enabled systems like *HearEase* not only revamps operations but also aligns local justice with national e-governance modernization efforts championed by the Department of the Interior and Local Government (DILG) [23].

Synthesis

Table 2: Synthesis

Enhancing Process Automation Using Reinforcement Learning and Deep Neural Networks	AI-based Smart Management Information System for Enterprise Judicial Cases for Power Grid Enterprises	Exploring the Applicability of Artificial Intelligence in Local Government Units	Readiness for AI Adoption of Philippine Business and Industry: The Government's Role in Fostering Innovation- and AI-Driven Industrial Development	Using Artificial Intelligence to Provide Intelligent Dispute Resolution Support	HearEase: Optimizing Barangay Hearing Scheduling and Predicting Case Resolution Times Using DNN (This Study)
Highlighted the "black box" nature of DNNs, noting how lack of transparency hinders stakeholder trust and complicates debugging.	Emphasized integrating legal and operational context to create a tailored system that addresses specific industry challenges.	Identified primary adoption barriers in LGUs: limited technical expertise, poor data infrastructure, and financial constraints.	Quantitatively assessed AI readiness across sectors using digital infrastructure indices to establish a baseline for adoption.	Reviewed the evolution of Online Dispute Resolution (ODR) to highlight gaps and the necessity of user-centric design.	Integrates AI predictive modeling with real-world barangay data to optimize hearing schedules and validate administrative efficiency.

Enhancing Process Automation Using Reinforcement Learning and Deep Neural Networks	AI-based Smart Management Information System for Enterprise Judicial Cases for Power Grid Enterprises	Exploring the Applicability of Artificial Intelligence in Local Government Units	Readiness for AI Adoption of Philippine Business and Industry: The Government's Role in Fostering Innovation- and AI-Driven Industrial Development	Using Artificial Intelligence to Provide Intelligent Dispute Resolution Support	HearEase: Optimizing Barangay Hearing Scheduling and Predicting Case Resolution Times Using DNN (This Study)
RL and DNN models struggle to generalize learned behaviors across different tasks and industries.	Used NLP, knowledge graphs, and deep learning to extract information and derive insights from judicial case documents.	Proposed capacity building, academic partnerships, and data infrastructure investments to address adoption challenges.	Case studies highlight how organizational size, sector, and geography influence AI integration success in PH firms.	Analyzed early ODR systems to show how prioritizing technology over user needs leads to design shortcomings.	Uses historical hearing records, case durations, and scheduling procedures from Barangay Tetuan as primary training data.
Emphasizes high-quality data; warns that biased	Uses active learning and human-machine labeling to	Highlights streamlined processes, chatbots,	Employed a framework to categorize obstacles: digital	Deconstructs ODR into essential elements	Follows structured preprocessing (cleaning/labelin

Enhancing Process Automation Using Reinforcement Learning and Deep Neural Networks	AI-based Smart Management Information System for Enterprise Judicial Cases for Power Grid Enterprises	Exploring the Applicability of Artificial Intelligence in Local Government Units	Readiness for AI Adoption of Philippine Business and Industry: The Government's Role in Fostering Innovation- and AI-Driven Industrial Development	Using Artificial Intelligence to Provide Intelligent Dispute Resolution Support	HearEase: Optimizing Barangay Hearing Scheduling and Predicting Case Resolution Times Using DNN (This Study)
datasets cause overfitting and poor real-world generalization .	improve accuracy; hybrid expert-system optimizes via user feedback.	data-driven resource allocation, and increased citizen engagement.	infrastructure, awareness, skills gaps, and funding.	(Triaging, Advisory, Drafting) to guide intelligent system design.	g) and rigorous DNN parameter tuning to predict resolution timelines.
High computational costs of training RL/DNN models create bottlenecks in	Synthesizes semantic retrieval with predictive modeling to recommend strategies	Highlights pilot projects (chatbots) and emphasizes ethical oversight to ensure	Interviewed diverse stakeholders (business/govt) to capture the full ecosystem of	Developed a conceptual user-centric ODR model integrating hybrid tools for	Assesses effectiveness via model accuracy, scheduling efficiency, and user satisfaction

Enhancing Process Automation Using Reinforcement Learning and Deep Neural Networks	AI-based Smart Management Information System for Enterprise Judicial Cases for Power Grid Enterprises	Exploring the Applicability of Artificial Intelligence in Local Government Units	Readiness for AI Adoption of Philippine Business and Industry: The Government's Role in Fostering Innovation- and AI-Driven Industrial Development	Using Artificial Intelligence to Provide Intelligent Dispute Resolution Support	HearEase: Optimizing Barangay Hearing Scheduling and Predicting Case Resolution Times Using DNN (This Study)
resource-limited environments.	based on historical legal events.	transparency and public trust.	AI adoption challenges.	diverse dispute scenarios.	vs. traditional methods.
Errors during the RL learning phase can have costly safety consequences in automation.	Integrated encryption and access controls to safeguard data integrity and confidentiality during processing.	Stresses multi-stakeholder collaboration and best-practice sharing to enhance confidence in AI integration.	Analyzed international case studies to identify strategies and policies adaptable to the Philippine context.	Initial metrics show that user-centric design significantly improves satisfaction and resolution rates.	Ensures data security for case records. Limitations include potential training bias and scalability needs.

Conceptual Framework

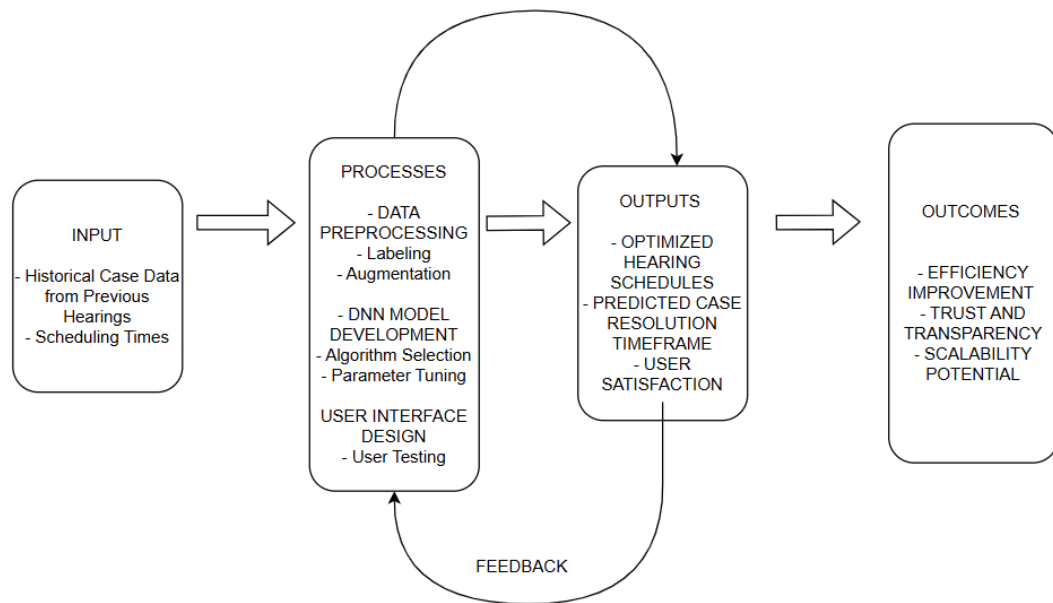


Figure 1: Conceptual Framework

The conceptual model shows how DNN will be used to maximize case scheduling and resolution in barangay hearings. Input, Processes, Outputs, and Outcomes form four main components of it. Historical case data—including past hearing records, case lengths, and scheduling procedures—makes up the Input section. The AI-driven system is built upon these databases.

The component on processes emphasizes the evolution of artificial intelligence systems. Data preparation starts it. Historical data is labeled and augmented to improve learning accuracy. After that, algorithm choice and parameter tuning help to build the Deep Neural Network (DNN) model so as to

optimize case predictions. Furthermore planned is a user interface that guarantees accessibility and usability by thorough testing.

The system generates expected case resolutions based on past data, idealized hearing schedules, and a user satisfaction assessment. For barangay officials, these outputs help to automate scheduling, improve case management, and sharpen decision-making. At last, the results of the framework highlight long-term advantages including better transparency, more case handling efficiency, and the possible system scalability to other barangays. The framework incorporates a feedback loop to enable ongoing development depending on user experience and fresh data.

CHAPTER III

METHODOLOGY

Research Design

This study utilizes a Developmental Research Design, specifically focusing on the creation, deployment, and evaluation of HearEase, an AI-powered hearing scheduling system. This design is appropriate as the study aims to produce a functional software artifact that addresses administrative inefficiencies through technological intervention. The system will be deployed as a Web-Based Application, ensuring accessibility for barangay officials and constituents.

To analyze the system's efficacy, the research employs a Quantitative approach. This involves two phases:

Model Evaluation: Assessing the DNN's performance using standard error metrics (MAE, RMSE, R^2 , POISSON DEVIANCE) and prediction accuracy against the historical and synthetic datasets from Barangay Tetuan.

System Evaluation: Conducting User Acceptance Testing (UAT) via ISO 25010 standards to measure usability, efficiency, and functionality based on feedback from the Lupong Tagapamayapa and barangay staff.

Participants and Sampling

The study will employ a purposive sampling method, as the participants will be selected based on their direct involvement and experience with the barangay justice system in the pilot site of Barangay Tetuan. The participants will consist of two groups:

1. Barangay Officials and Lupon Tagapamayapa (Expert Users)

This group comprises the primary administrators of the system, including the Punong Barangay, Barangay Secretary, and the appointed Lupon members. The study aims for Total Enumeration or the inclusion of the entire available workforce responsible for mediation, estimated at 10 to 15 participants. This comprehensive coverage ensures that the feedback represents the collective operational needs of the barangay, validating the system's functional suitability and workflow efficiency.

2. Local Residents (End-Users)

To evaluate the public-facing portal, 25 residents who have prior experience with filing complaints or attending hearings will be selected. These participants will be engaged for User Acceptance Testing (UAT) to assess the system's usability, interface clarity, and transparency. This sample size is sufficient to identify usability flaws and gauge general user satisfaction in a developmental pilot study context.

Data Source

The research data for the purpose of developing an AI technology system using Deep Neural Networks (DNNs) to enhance barangay hearings will be from primary data and secondary data. The historical case records will present a vital component as they will be used to obtain information on previous barangay cases. Case records will include details like types of cases, duration of cases, predicted number of hearings, resolutions of cases, and any delays observed in case processing. These records will be provided by the records management division of Barangay Tetuan, Zamboanga City.

Demographic data on the parties in the cases will also be collected including persons' ages, gender, and other socioeconomic status information. This will be done to identify trends in handling cases, along with identifying user needs. Demographic data will be collected in coordination with barangay officials and may also emerge from municipality census files.

The study will also acquire qualitative and perception data through the following means:

Surveys issued to members of the Lupong Tagapamayapa and barangay workers to further understand current practices, perceived challenges, and opportunities for enhanced case management of semi-structured interviews of barangay officials to gather deeper understandings of operational workflows and trends in case thinking. Then, all data collected will be compiled and

pre-processed, prior for analysis. This means structuring historical case data, anonymizing sensitive information, coder survey, and interview responses into datasets for analysis.

Lastly, User Acceptance testing, UAT will occur during the system evaluation process. Structured feedback and surveys will be used by users to assess usability in terms of the effectiveness of the system and user satisfaction. The UAT feedback will be an important input information for system refinement and evaluating the system's effectiveness on the case management process.

Through this empirical approach to data gathering means that the development of the system is driven by real-world data, contextual reality of the human factors operationalized, and the input feedback from the stakeholders engaged.

Data Gathering Instrument

The core dataset for training the Deep Neural Network (DNN) model is derived from the actual case blotters of Barangay Tetuan, Zamboanga City, covering the period from 2018 to 2025. This timeframe captures pre-pandemic, lockdown, and post-pandemic operational contexts.

To understand the nuances of the Katarungang Pambarangay workflow, semi-structured interviews will be conducted with the Punong Barangay, Barangay Secretary, and Lupon members. These interviews focused on

identifying unwritten rules of scheduling, common causes of delays, and the impact of party relationships on settlement speed. Direct observations of the hearing scheduling process were also performed to map the manual workflow and identify bottlenecks.

For system evaluation, perception data will be gathered through UAT surveys administered to both Lupong Tagapamayapa Secretary and selected residents. These surveys utilize standard usability metrics (ISO 25010) to assess the system's functional suitability, ease of use, and efficiency in comparison to traditional manual procedures.

All collected data adheres strictly to the Data Privacy Act of 2012 (RA 10173). Historical records will be anonymized during the digitization process; names of actual constituents will be replaced with synthetic identifiers or generic placeholders to protect confidentiality. No sensitive socioeconomic or financial data beyond what is necessary for case classification will be recorded.

Data Gathering Technique and Procedures

Data collection for this study will employ a variety of methods that aim to effectively gather and collect qualitative and quantitative data. These methods are:

- Surveys to document resident and legal staff comments, opinions, and impressions of their barangay hearing experiences.

- Semi-structured interviews allow one to have in-depth conversations with the barangay official in which open-ended questions let them address operational concerns, and experiences running the hearings.
- Data extraction to compile past case files including specifics on case types, length, resolutions, schedules and results.
- Observational studies to directly view barangay hearings and acquire first understanding of the procedural dynamics and interactions during the events.

Using these approaches requires methodical steps as follows:

- Step 1 - Get permission from Barangay Tetuan officials pertaining to the study of past case records and surveys. This meant personally getting in touch with officials to let them know about the goals of the study and the need for their cooperation.
- Step 2 - Create and distribute a structured survey aimed at residents and legal staff members. The survey should consist of a mix of closed and open-ended questions to gain quantitative data and qualitative feedback on their experiences with the current barangay hearing process.
- Step 3 - Conduct semi-structured interviews with barangay officials. This may involve guided questions while also allowing space for officials to contribute additional comments about their operating challenges and perceived effectiveness of the current process.

- Step 4 - Obtain historical case data through barangay record cases. This step will involve a review of these case files with barangay staff to uncover relevant information on case type, timeframes, outcomes, and scheduling.
- Step 5 - Observe barangay hearings as a way to gain qualitatively representative data on procedural dynamics, interaction patterns, and operational challenges described in the surveys and interviews. This data will complement the overall understanding of the barangay hearing processes.
- Step 6 - Aggregate and organize all data that has been collected with pre-processing. This will involve data cleaning for error reduction, coding survey responses, and reconstructing data as datasets ready for analysis and model development.

Data Analysis

The data gathered through the aforementioned techniques will be analyzed using a Convergent Parallel Mixed Methods approach, integrating both quantitative and qualitative findings to provide a holistic assessment of the HearEase system.

1. Data Pre-processing and Cleaning

Prior to analysis, the historical case records will undergo rigorous pre-processing. This involves Data Cleaning to remove duplicate entries or formatting errors, Normalization to scale numerical values, and One-Hot

Encoding to convert categorical variables into machine-readable formats. The dataset will be split into training (80%) and validation (20%) sets to prepare for the Deep Neural Network (DNN) training.

2. Quantitative Analysis (AI Model Performance)

To evaluate the effectiveness of the AI prediction engine, the study will utilize standard regression error metrics rather than simple accuracy percentages. The following metrics will be calculated:

- **Mean Absolute Error (MAE)** - To measure the average magnitude of errors in predicting the number of hearings and resolution days.
- **Root Mean Square Error (RMSE)** - To penalize larger prediction errors and ensure reliability in complex cases.
- **Mean Poisson Deviance:** Specifically utilized to evaluate the Hearings Head of the neural network. Since the number of mediation sessions is count data (non-negative integers), Poisson Deviance provides a more accurate measure of the model's "goodness-of-fit" than standard squared error metrics. A lower deviance value indicates a superior fit for the discrete distribution of barangay hearings.
- **Coefficient of Determination:** Used to quantify the proportion of variance in the targets that is predictable from the input features. This metric validates the "explanatory power" of the socio-legal variables (Case Type, Relationship, and Severity) on the final resolution timeline.

- **Descriptive Statistics:** Measures such as Mean, Median, and Standard Deviation were used to summarize the distribution of the 1,199-record dataset, ensuring that the stratified sampling maintained the natural "Severity Gradient" and "Temporal Patterns" (e.g., COVID-era delays) of the original records.

3. Qualitative Analysis (Interviews and Observations)

Qualitative data gathered from semi-structured interviews with the Lupong Tagapamayapa and operational observations will be subjected to Thematic Analysis. This process involves coding responses to identify recurring patterns regarding scheduling bottlenecks, administrative pain points, and user readiness for digital adoption. These insights will validate the necessity of specific system features.

4. System Evaluation (User Acceptance Testing)

Data from the User Acceptance Testing (UAT) surveys will be analyzed using Descriptive Statistics. The responses, rated on a Likert Scale, will be computed for Weighted Means to determine the system's performance in terms of Functional Suitability, Performance Efficiency, and Usability, adhering to ISO 25010 Software Quality Standards.

5. Comparative Analysis

Finally, the study will perform a comparative analysis between the traditional manual method and the HearEase automated system. Key performance indicators (KPIs), specifically Scheduling Time (minutes) and Conflict Detection Rate, will be compared to statistically quantify the efficiency gains provided by the AI system. Why

Software Development Methodology

This study employs the Agile Software Development Life Cycle (SDLC) to ensure the efficient design, development, and deployment of HearEase. The Agile methodology was selected for its iterative nature, allowing for continuous refinement based on user feedback and the dynamic operational requirements of the barangay justice system.

By utilizing the Scrum framework within Agile, the development process is divided into sprint cycles. This approach enables the researchers to address functional issues early, adapt to the specific needs of the Lupong Tagapamayapa, and ensure that the final output is a robust, user-centric administrative and judicial support system. Continuous integration and testing are embedded in the workflow to guarantee system stability and data integrity.

Each sprint cycle consists of the following phases:

- **Planning:** Defining the backlog, user stories, and specific objectives for the sprint based on barangay procedural rules.

- **Design:** creating process flowcharts, entity-relationship diagrams (ERD), and high-fidelity wireframes to visualize the system architecture.
- **Development:** Writing the source code for the frontend interfaces and backend logic, including the integration of the AI prediction engine and scheduling algorithms.
- **Testing:** Conducting Unit Testing and Integration Testing to ensure all modules function correctly and securely.
- **Review:** Demonstrating the developed features to Barangay Tetuan officials to gather immediate feedback and validation. **Deployment:** Rolling out the approved iteration to the live web server for real-world simulation and user acceptance.

Optimization of Hearing Scheduling

To optimize the scheduling workflow within HearEase, the system employs a Hybrid Scheduling Engine that combines AI-driven predictive analytics with a rule-based algorithm. This approach ensures that hearing timelines are not only historically accurate but also logistically feasible within the barangay's resource constraints.

1. Predictive Resource Allocation (AI Role)

The core optimization begins with the Deep Neural Network (DNN) Prediction Engine. Upon case filing, the AI analyzes parameters such as case severity, party complexity, and historical resolution patterns to forecast the Total

Number of Hearings required. This effectively estimates the "volume" of the case, allowing the system to reserve the necessary number of time slots in advance, rather than scheduling strictly on a per-hearing basis.

2. Conflict Detection and Slot Allocation (Algorithm Role)

Once the required number of hearings is predicted, the system's Calendar Optimization Algorithm calculates specific dates and times. It utilizes a Conflict Detection Mechanism to scan the database for existing appointments, holidays, and venue availability. If a projected date falls on a non-working day or a fully booked slot, the system automatically shifts the schedule to the next available working day, ensuring zero overlap and maximizing venue utilization.

3. Adaptive Scheduling Intervals (Expedite Feature)

Recognizing that high-stakes cases require swifter resolution, the system includes an Adaptive Interval Feature. While standard cases follow the default 7-day interval (e.g., Monday-to-Monday) to allow for summons delivery, the system allows Administrators to toggle an "Expedite Mode" for urgent cases. This overrides the standard interval with a shortened duration (e.g., 2–3 days), triggering the algorithm to find the earliest available slots while maintaining conflict checks.

4. Automated Document Synthesis

Database Design



Figure 2: HearEase Relational Database Schema highlighting AI Model Tracking and Multi-Platform Integration.

HearEase utilizes Supabase, a cloud-native platform, to manage its PostgreSQL-based relational database. This selection provides advanced features such as Row-Level Security (RLS) for granular data privacy, real-time data broadcasting, and native support for complex JSONB data types (used for flexible scheduling and generated document data). The schema is designed to handle multi-party case relationships, automated hearing scheduling, and the logging of AI model performance metrics.

The database is organized into five core functional modules:

1. **Case Management Module:** Centered around the `cases_case` table, this module manages the lifecycle of a complaint. It links to `cases_casetype` (severity and classification), `cases_settlementtype`, and uses junction tables (`cases_case_complainants` and `cases_case_respondents`) to handle multiple parties involved in a single dispute.
2. **Hearing & Scheduling Module:** This module manages the logistical execution of the Katarungang Pambarangay process. The `hearings_hearing` table tracks dates, times, and statuses, while `hearings_hearingattendance` records the presence of parties. It also integrates `hearings_nonworkingday` to ensure the scheduling algorithm avoids holidays and weekends.

3. **User & Security Module:** Beyond basic authentication, this module includes `users_notificationpreference` to manage SMS and Email alerts via the custom gateway and Resend API. It also utilizes `users_otp` for secure multi-factor verification during sensitive administrative actions.
4. **AI & Analytics Module:** Unique to this study, the `AIModel_traininghistory` and `AIModel_retrainconfig` tables track the performance of the Deep Neural Network. They store metrics such as Mean Absolute Percentage Error (MAPE) for both days and hearings, allowing the system to monitor prediction accuracy over time.
5. **Integration & Document Module:** This manages external synchronization with Google Calendar via OAuth tokens (`google_calendar_googlecalendartoken`) and stores the templates and JSON data for the Automated Document Synthesis engine (`documents_generateddocument`).

To comply with the Data Privacy Act of 2012 (RA 10173), the database utilizes Supabase Row-Level Security (RLS) policies. This ensures that sensitive personal information in the `case_persons_caseperson` table is only accessible to authorized `is_admin` or `is_superuser` users. Furthermore, all personally identifiable information (PII) is isolated from the AI training datasets to prevent data leakage during model optimization."

Development

The development of HearEase adheres to the Agile methodology, structured into four distinct technical phases:

1. Frontend Development

- The client-side interface is constructed using a modern technology stack comprising Tailwind CSS for utility-first styling and shadcn/ui for accessible, high-quality UI components. This combination ensures a responsive, professional design that adapts to both desktop monitors and mobile devices. Visual communication is enhanced through the integration of Lucide Icons, providing intuitive cues for non-technical users. FullCalendar, a robust JavaScript scheduling library, is utilized to handle the complex logic required for rendering hearing dates and interactive calendar views.

2. Backend Development

The backend of the system was developed using Django REST Framework (DRF), an extension of the Django web framework designed for building Web APIs. The primary factor for choosing DRF was its ability to provide a robust, scalable, and secure interface between the PostgreSQL database and

the frontend application. This setup eliminates the overhead of a traditional monolithic server, providing a faster and more scalable user experience.

- **Data Management (CRUD):** The backend executes secure Create, Read, Update, and Delete operations for case records, enforcing referential integrity within the database.
- **Hybrid Scheduling Engine:** This is the core logical module. It acts as the bridge between the AI and the calendar. Once the AI predicts the number of hearings, this Django module calculates the specific dates (using the 7-day or Expedited interval), performs Conflict Detection against existing appointments, and finalizes the schedule.
- **API Architecture:** Custom RESTful API endpoints manage the communication between the frontend and the AI model, ensuring low-latency data transfer during prediction requests.
- **Security & Authentication:** Firebase Authentication handles session management, supporting secure email/password and OAuth protocols. Data security is further reinforced at the database level through Supabase Row-Level Security (RLS), ensuring that sensitive case blotters are only accessible to authorized barangay officials. Communication security is maintained via SSL/TLS encryption across all API requests.
- **Automated Notification Alerts:** To facilitate timely information dissemination, the system integrates the Resend API for transactional emails (account verifications and system alerts) and a Custom Android

SMS Gateway. The gateway utilizes a local Android device as a digital bridge, allowing the system to send SMS notifications through local cellular networks, ensuring high deliverability and cost-efficiency for the barangay.

3. AI Model Development

- **Data Preprocessing:** The input layer processes a 30-dimensional feature vector. This is derived through a multi-step preprocessing pipeline:
 1. **Ordinal Encoding:** Case Severity is mapped to an integer scale (Low: 1, Medium: 2, High: 3).
 2. **One-Hot Encoding:** Categorical variables including Case Type (15 types), Relationship (6 types), and Lockdown Status (3 types) are transformed into 27 binary columns.
 3. **Z-Score Normalization:** Numerical features such as the number of complainants and respondents are standardized to a mean of 0 and standard deviation of 1 using StandardScaler.
- **Feature Engineering:** To optimize prediction accuracy, the model utilizes specific engineered features derived from the "Barangay Logic":

- **Party Complexity:** The count of Complainants and Respondents is factored in to predict scheduling delays caused by availability conflicts.
 - **Relational Dynamics:** The relationship between parties (e.g., Family vs. Stranger) is weighted to account for the emotional intensity of disputes.
 - **Contextual Factors:** Variables such as Intoxication or Recidivism are included to refine the estimated resolution timeline.
- **Model Architecture:**

The HearEase system utilizes a Multi-Output Multilayer Perceptron (MLP) architecture built with TensorFlow 2.x and Keras. Unlike standard single-task models, this architecture employs a shared feature extraction trunk (comprising 128 and 64-unit Dense layers with ReLU activation) that branches into two independent prediction heads.

This 'split-head' design allows the model to learn common socio-legal patterns in the shared trunk while specializing the output layers for two distinct distributions:

1. **Hearings Head:** Predicts a discrete count (1–6 sessions).

2. **Resolution Head:** Predicts a continuous timeframe (3–172 days). To prevent overfitting on the 1,199-record dataset, a Dropout layer (rate=0.2) is integrated into the shared trunk, ensuring the model generalizes well to new cases rather than memorizing training data.

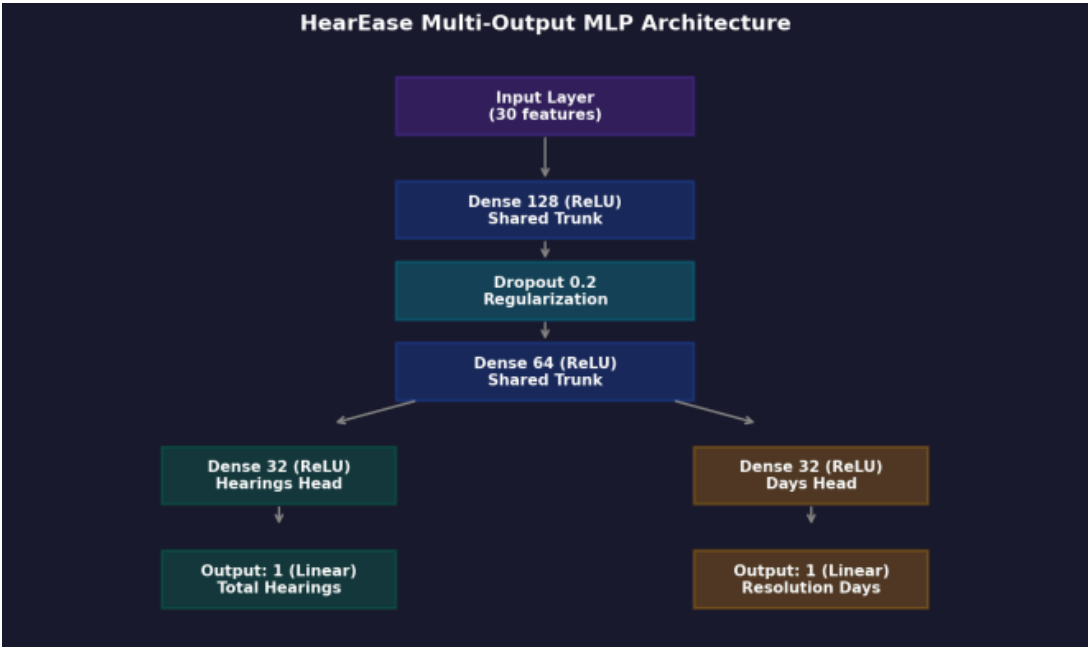


Figure 3: Model Architecture

Table 3: Architecture Breakdown

Layer	Type	Units	Activation	Purpose
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input_features	Input	30	—	Receives preprocessed feature vector
shared_dense_128	Dense	128	ReLU	Shared feature extraction
shared_dropout	Dropout	20%	—	Regularization to prevent overfitting
shared_dense_64	Dense	64	ReLU	Shared deeper representation
hearings_dense_32	Dense	32	ReLU	Hearings-specific features
hearings_output	Dense	1	Linear	Predicts total hearings count
days_dense_32	Dense	32	ReLU	Days-specific features
days_output	Dense	1	Linear	Predicts resolution time in days

Table 4: Core ML Stack

Library	Role
TensorFlow 2.x	Deep learning framework — model building, training, inference
Keras (bundled with TF)	High-level API for defining layers, models, and training loops
scikit-learn	Preprocessing (OneHotEncoder, StandardScaler), metrics, train/test splitting
NumPy	Numerical operations, array manipulation, random seeding
pandas	CSV loading, DataFrame operations, feature manipulation

Table 5: Preprocessing Libraries

Library Component	Role
sklearn.preprocessing.OneHotEncoder	Converts categorical features → binary columns (27 total)
sklearn.preprocessing.StandardScaler	Z-score normalizes numerical features to mean=0, std=1
sklearn.model_selection.train_test_split	80/20 train/test split with random_state=42

Table 6: Evaluation Metrics Libraries

Function	Role
mean_absolute_error	Average absolute prediction error (MAE)
mean_squared_error	Squared error for RMSE calculation
r2_score	Coefficient of determination (R^2)
mean_poisson_deviance	Tailored metric for count data accuracy

Table 7: Utility Libraries

Library	Role
joblib	Serialization of encoder, scaler, feature_info to .pkl files
matplotlib	Graph generation for training curves and evaluation plots
json	Saving/loading training results for persistence
base64 / io	Converting matplotlib figures to base64 PNGs for frontend

4. Testing & Deployment

- Testing Strategy: The system undergoes a tiered testing approach:
 - Unit Testing: To validate individual functions (e.g., date calculation logic).
 - Integration Testing: To ensure seamless data flow between the Database, AI Model, and User Interface.
 - User Acceptance Testing (UAT): Conducted with Lupon Tagapamayapa Secretary using ISO 25010 criteria to validate functional suitability and usability.
 - Cloud Deployment: Following successful testing, HearEase will be deployed on a secure cloud hosting platform. This architecture ensures 24/7 accessibility for residents tracking their cases while maintaining a centralized, secure database for barangay administrators.
 - Security & Privacy: Communications are secured via SSL/TLS encryption (HTTPS). To comply with the Data Privacy Act of 2012, all public-facing data is anonymized, and personal records are accessible only via encrypted authentication.

Developmental Tools and Cost

Table 8: Developmental Tools and Cost

Name	Purpose	Price	Quantity	Total
Programming Languages	For developing the system and the DNN Model	FREE	-	FREE
Libraries: Scikit-learn, TensorFlow/Keras	Machine Learning Algorithms	FREE	-	FREE
Visual Studio Code	Integrated development	FREE	-	FREE
Google Cloud Platform (GCP) or AWS (Free Tier)	Cloud Computing (Model Training, Data Storage)	FREE	1	FREE

Name	Purpose	Price	Quantity	Total
Cloud Hosting (Beyond Free Tier)	Potential for increased usage beyond free limits	₱1,000.00	-	₱1,000.00
Survey Platform (Printed)	Data Gathering from Barangay Officials, Residents	₱2.00	50	₱100.00
Statistical Software (Jamovi)	Data analysis and testing of collected data	FREE	1	FREE
Online Collaboration Tools (Notion)	Communication, Project Management	FREE	-	FREE
Contingency/Miscellaneous Expenses (Internet, Fare, etc.)	Unexpected Costs During Development and Testing Phases	₱2,000.00	-	₱2,000.00

Name	Purpose	Price	Quantity	Total
PyInstaller	Packaging Python-based HearEase into an executable	FREE	-	FREE
Inno Setup (Windows)	Installer for Windows	FREE	-	FREE
Grand Total				₱3,050.00

Evaluation

The evaluation phase synthesizes the quantitative and qualitative data gathered to assess the practical value and efficacy of the HearEase system. This process is designed to validate whether the specific research objectives have been met, focusing on technical accuracy, user acceptance, and operational impact.

1. Evaluation of AI Prediction Accuracy

To verify the system's ability to forecast resolution timelines, dataset will be used for testing. The model's predictions will be compared against the actual historical values using statistical error metrics:

- Mean Absolute Error (MAE): Utilized to quantify the average deviation in the predicted number of days and hearing counts. This provides a clear, real-world interpretation of the error (e.g., an MAE of 0.70 hearings indicates the prediction is usually within one session of the actual result).
- Mean Poisson Deviance: Specifically employed to evaluate the "Total Hearings" prediction head. Since mediation sessions are discrete count data (non-negative integers), Poisson Deviance provides a more accurate measure of the model's goodness-of-fit than standard regression metrics, identifying how well the model captures the frequency distribution of barangay cases.
- Coefficient of Determination: Used to assess the model's explanatory power. This metric measures the proportion of variance in resolution time that is predictable from the input features (Severity, Relationship, and Case Type), proving that the chosen variables are statistically significant drivers of case longevity.

Accuracy Rate (Margin-based): To measure the percentage of predictions that fall within a practically acceptable margin. For Resolution Time, a margin of hearing was applied. High success rates in these margins validate the reliability of the "Predictive Resource Allocation" feature for administrative use.

2. Assessment of User Acceptance

Data from the surveys administered to the Lupong Tagapamayapa and residents will be analyzed using Descriptive Statistics. User feedback will be quantified based on the ISO 25010 Software Quality Standards, specifically measuring:

- **Functional Suitability:** Does the system perform the scheduling tasks as required?
- **Usability:** Is the interface intuitive for non-technical barangay officials?
- High weighted mean scores in these categories will serve as the indicator of the system's successful design and user-friendliness.

4. Data Triangulation

To ensure the reliability of the findings, the study will employ Data Triangulation. Consistency will be cross-verified between the Quantitative Results (System Logs/AI Metrics) and the Qualitative Insights (Interview Feedback). For instance, if the AI reports a 50% reduction in scheduling time, this should correspond with interview statements from the Barangay Secretary regarding reduced workload. This holistic approach ensures that the evaluation captures both the technical performance and the lived experience of the stakeholders.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the findings of the study, organized according to the specific objectives defined in Chapter I. It covers the analysis of current workflows, the results of the data collection and AI training, the features of the developed system, and the evaluation of its predictive accuracy and usability.

4.1 Analysis of Existing Case Workflows

To identify requirements for automation, the researchers conducted interviews and observations at Barangay Tetuan.

- **Workflow Constraints:** The existing process relies on a manual logbook system. Scheduling is done on an ad-hoc basis, usually set exactly 7 days apart without considering case complexity.
- **Bottlenecks:** Key issues identified include overlapping appointments (conflicts), difficulty in tracking case history, and inconsistent estimation of how many hearings a case might require.
- **Requirement for AI:** The findings confirmed that a "one-size-fits-all" 7-day schedule does not account for high-severity cases or complex relationships, justifying the need for a predictive model.

4.2 Data Collection and Dataset Characteristics

The researchers collected and processed 1,199 stratified records from the Barangay Tetuan case blotters (2018–2025).

- **Case Distribution:** The dataset reflects 15 distinct case types. As shown in the findings, Gossip/Oral Defamation (14.0%) and Neighborhood Disturbance (12.7%) are the most common.
- **Validation of Socio-Legal Patterns:** Analysis of the collected data revealed a Severity Gradient: Low-severity cases average 2.75 hearings, while High-severity cases average 4.05 hearings. Furthermore, Family disputes were found to be the most complex, requiring a mean of 4.06 hearings.
- **Data Integrity:** The stratified sampling preserved the sharp drop in case volume during 2020–2021, representing the COVID-19 lockdown impact, which was critical for training the AI to distinguish between case complexity and administrative delays.

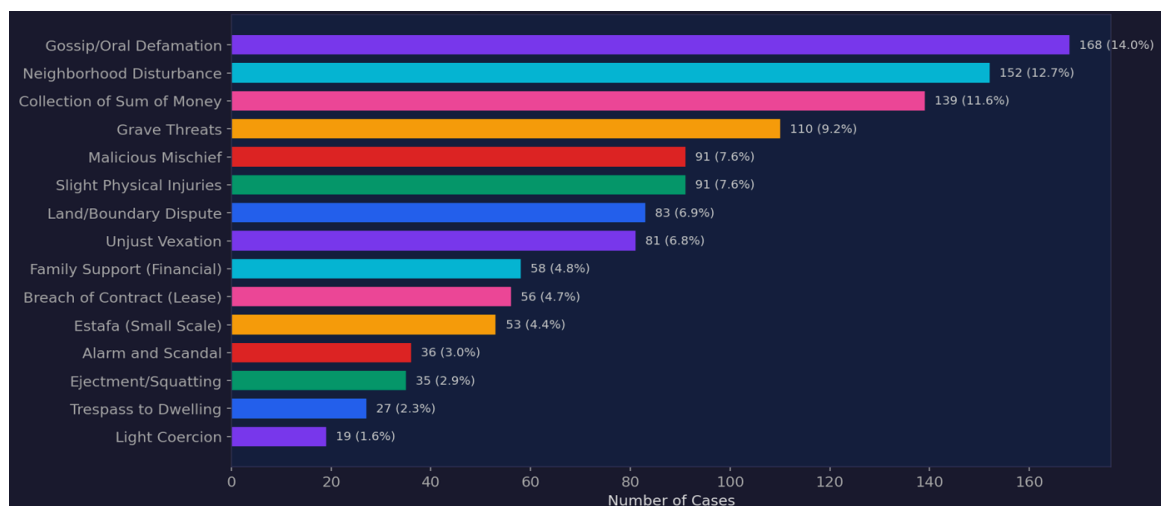


Figure 4: Case Type Distribution

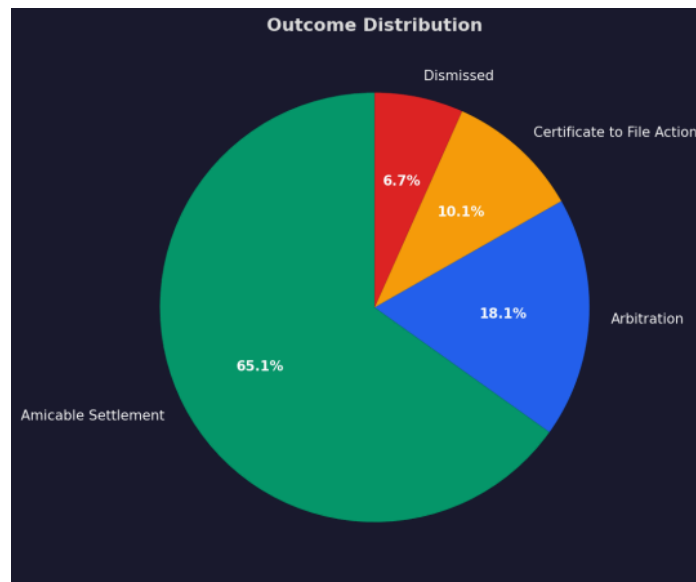


Figure 5: Outcome Distribution

4.3 Development of HearEase

The system was successfully developed using a cloud-native architecture.

Tech Stack Implementation: Unlike traditional monolithic designs, HearEase utilizes Supabase (PostgreSQL) for relational data and Firebase for real-time

authentication. The frontend was built with Tailwind CSS and shadcn/ui for accessibility.

- **AI-Driven Features:** The system integrates a Multi-Output Multilayer Perceptron (MLP). When a case is filed, the AI predicts the number of hearings and resolution days. These values are then passed to the FullCalendar scheduling engine.
- **Communication Bridge:** To address Objective 1's notification bottleneck, a Custom Android SMS Gateway was integrated to provide instant alerts to residents, ensuring procedural adherence even for users without smartphones.

4.4 Evaluation of Predictive Accuracy

This summary integrates the analysis of the dataset's framework, detailing how every core variable affects a case's complexity (Total Hearings) and duration (Resolution Time), with a specific focus on the provided "Summary Table for Dataset."

I. The Core Dichotomy: Hearings vs. Resolution Time

The dataset reveals a fundamental relationship between the number of hearings and the total days to resolve a case:

- **Normal Operational Context (2018–2019, 2022–2025):** There is a direct positive correlation. Since hearings are scheduled strictly one week apart, Resolution Time is largely a function of Total Hearings $\times$ 7 Days.
- **Pandemic Context (Lockdown Status):** The correlation decouples. During ECQ (Strict Lockdown), Resolution Time increases massively (+90 to +150 days) regardless of the number of hearings. This confirms that the AI model successfully learned to distinguish between case difficulty and force majeure delays.

Table 9: Hearings x Resolution Times

Factor	Total Hearings (Longevity)	Resolution Time (Days)	Key Explanation
Normal Environment	Direct Driver	Direct Multiplier ($\approx$ Total Hearings $\times$ 7 days)	A strong positive correlation exists under standard conditions.
Lockdown Status	Neutral	Massive Increase ($+90$ to $+150$)	External administrative freeze decouples the metrics, proving the delay is not due to case difficulty.

		days)	
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II. Internal Factors That Lengthen Case Duration (The Multipliers)

The Deep Neural Network (DNN) identified specific features that significantly lengthen or shorten the judicial process:

Table 10: Internal Factors That Lengthen Case Duration

Variable	Correlation to Hearings	Explanation from Framework / Table	Effect on Number of Hearings	Effect on Resolution Time (Days)
Case Type	Positive	"Have inherently different complexities, essentially determines the baseline difficulty of a case,"	Increases (Complex cases increases the number of hearings needed to be settled while simple cases usually take lesser time to be resolved)	Lengthens (up to 4x difference between complex and simple cases)
Case Severity	Positive	"Is subjective to the person assessing the case, but it tells a rough idea on how urgent or difficult the case is"	Increases (High severity = 4 avg vs Low severity = 2.75 avg hearings)	Lengthens (Due to increased hearings and resistance)
Number of Respondents	Positive	"More people = more scheduling conflicts and conflicting testimonies."	Slightly Increases (More parties to mediate and schedule)	Lengthens (Directly impacts scheduling and "Turuan")
Recidivism (Prev. Cases)	Positive	"Habitual offenders are resistant to amicable settlement."	Increases (Pushes case toward maximum hearings/CFA)	Lengthens (Pushes case towards maximum hearings/CFA)

Family Relationship	Positive	"Emotional baggage extends mediation time."	Increases (More sessions needed for emotional mediation)	Lengthens (Requires more time for emotional venting, thus more hearings)
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III. Internal Factors That Lessen Case Duration (The Shortener)

Table 11: Internal Factors That Shortens Case Duration (The Multipliers)

Variable	Correlation to Hearings	Explanation from Framework / Table	Effect on Number of Hearings	Effect on Resolution Time (Days)
Intoxication (at incident)	Negative	"Sobering up" leads to faster admission of guilt and apology.	Decreases (Respondent sobers up, admits guilt faster due to evidence of intoxication, fewer mediation sessions needed)	Shortens (Faster Resolution, fewer hearings)

IV. Case Type Patterns and Predicted Outcome (Settlement Type)

The case types are categorized by their predicted longevity, which influences the final Outcome (Settlement Type).

A. Low Longevity Cases (High Likelihood of Amicable Settlement)

These cases tend to resolve quickly, often in 1-2 hearings.

Table 12: Low Longevity Cases

Case Type	Percentage of Caseload	Pattern & Predicted Outcome
------------------	-------------------------------	--

Gossip/Oral Defamation	14.00%	"Heat of the moment" issue. High likelihood of Amicable Settlement with apology/retraction.
Neighborhood Disturbance	12.67%	Generally transactional, outcome depends on dispute nature (similar to Strangers/Neighbors).
Slight Physical Injuries	7.67%	Outcome depends on compensation for damages/medical expenses. Likely Amicable Settlement if resolved quickly.
Malicious Mischief	7.60%	"Petty Mischief." Likely Amicable Settlement with payment of damage.
Unjust Vexation	6.73%	Outcome depends on apology and cessation of activity. Likely Amicable Settlement .
Alarm and Scandal	3.00%	"Heat of the moment" issue. High likelihood of Amicable Settlement .
Ejectment/Squatting	2.93%	Although High Longevity due to procedure, if resolved, it's a formal settlement. If not, leads to CFA.
Trespass to Dwelling	2.27%	Outcome depends on severity. Can lead to Amicable Settlement or CFA.
Light Coercion	1.60%	Outcome depends on respondent's resistance to avoiding court record. Can lead to Amicable Settlement or CFA.

B. High Longevity Cases (Higher Likelihood of Difficult Settlement or CFA)

These cases demand more procedural steps or are driven by deep-seated issues, increasing hearings and resolution time.

Table 13: Internal Factors That Lengthen Case Duration

Case Type	Percentage of Caseload	Pattern & Predicted Outcome
Collection of Sum of Money	11.60%	High Longevity due to financial negotiations . Higher risk of CFA if repayment is not agreed upon.
Grave Threats	9.20%	High Severity. Outcome heavily influenced by Recidivism and Intoxication . Higher risk of CFA if respondent is resistant.
Land/Boundary Dispute	6.87%	High Longevity due to ocular inspection/document verification . Can lead to CFA if no agreement on boundaries.
Family Support (Financial)	4.80%	High Longevity due to financial negotiations . Outcome depends on ability to pay; can lead to CFA.
Breach of Contract (Lease)	4.67%	Outcome depends on ability to pay/fulfill contract. Can lead to CFA if not resolved.
Estafa (Small Scale)	4.40%	High Severity/Financial. Outcome depends on restitution. Higher risk of CFA if respondent is resistant or a recidivist.
Ejectment/Squatting	2.93%	(Reiterated from Low Longevity due to Outcome Nuance) High Longevity due to procedure. If unresolved, leads to CFA.

V. Detailed Summary of Important Predictors for Action

The analysis highlights key factors that significantly influence case duration and complexity, informing administrative decisions and predictive modeling.

1. Recidivism:

- **Correlation:** Positive to Hearings and Resolution Time.
- **Significance:** This is the strongest predictor of a failed amicable settlement, leading to cases being pushed to the maximum allowed hearings and time, ultimately resulting in a Certificate to File Action (CFA). It significantly lengthens Resolution Time.

2. Lockdown Status (Pandemic Era):

- **Correlation:** Neutral to Hearings, but Massive Positive to Days.
- **Significance:** This factor is crucial for data integrity and AI training. It allows for the decoupling of administrative delays from actual case complexity. A case taking a long time during a lockdown is due to external factors (Force Majeure), not inherent difficulty.

3. Number of Respondents:

- **Correlation:** Positive to Hearings and Resolution Time.
- **Significance:** Identified as the "single biggest administrative factor" due to scheduling conflicts and the need to resolve the "Blame Game" (Turuan). This directly impacts the ease of scheduling and the total duration of the mediation process.

4. Case Type (High Longevity Groups):

- **Correlation:** Positive to Hearings and Resolution Time.
- **Significance:** The 26.2%\mathbf{26.2\%}26.2% of cases in the High Longevity group (e.g., Land Disputes, Family Support, Collection of Money) are the primary drivers of internal case complexity. They demand specific procedural steps that inherently take more time and hearings, thus influencing the Resolution Time.

5. Family Relationship:

- **Correlation:** Positive to Hearings and Resolution Time.
- **Significance:** This "Emotional Factor" adds time because resolving deep-seated grudges and past history requires significant **emotional venting** across multiple hearings, thus extending Resolution Time.

6. Intoxication:

- **Correlation:** Negative to Hearings and Resolution Time.
- **Significance:** This is the key factor that can actively shorten resolution time. The "Sober Apology" mechanism leads to faster admissions and settlements, reducing both hearings and overall duration.

7. Case Severity:

- **Correlation:** Positive to Hearings and Resolution Time.
- **Significance:** The underlying threat of heavier court penalties motivates respondents to resist settlement and extend the process

to avoid legal records, thereby increasing hearings and resolution time.

4.4.1 Feature Engineering & Preprocessing

Step 1: Severity Encoding

Severity strings mapped to ordinal integers:

{"Low": 1, "Medium": 2, "High": 3}

Step 2: Categorical One-Hot Encoding

Feature	Categories	One-Hot Columns
Case_Type	15	15 binary columns
Settlement_Type (Outcome)	4	4 binary columns
Relationship	6	6 binary columns
Lockdown_Status	3	3 binary columns
Total	—	27 columns

Encoder uses `handle_unknown="ignore"` so unseen categories produce all-zero vectors.

Step 3: Numerical Standardization

Three numerical features Z-score normalized: $z = (x - \text{mean}) / \text{std}$

Feature	Raw Range	After Scaling
Severity	1–3	mean $\approx$ 0, std $\approx$ 1
Num_Complainants	~	mean $\approx$ 0, std $\approx$ 1
Num_Respondents	~	mean $\approx$ 0, std $\approx$ 1

Final Input Vector

The final input is a 30-dimensional vector: 27 one-hot + 3 standardized numerical features.

Training Pipeline

Data Split

Total: 1,199 samples

- Training: 80% $\rightarrow$ 959 samples

└─ Fit: 767 samples

└─ Validation: 192 samples

- Test: 20% → 240 samples (held out)

Table 14: Training Configuration

Parameter	Value	Rationale
Epochs	50	Sufficient for convergence
Batch Size	32	Standard for small-to-medium datasets
Validation Split	0.2	20% held for monitoring
Optimizer	Adam (lr=0.001)	Adaptive learning rate
Loss Function	MSE	Standard regression loss
Random Seed	42	Reproducible results

MSE penalizes large errors quadratically. MAE Loss gives equal weight to all errors. Poisson Loss could work for counts but MSE proved more stable. Huber Loss adds unnecessary complexity.

Table 15: Saved Artifacts

File	Format	Content
------	--------	---------

model.keras	Keras SavedModel	Full neural network weights and architecture
encoder.pkl	Joblib pickle	Fitted OneHotEncoder with category mappings
scaler.pkl	Joblib pickle	Fitted StandardScaler with mean/std values
feature_info.pkl	Joblib pickle	Column names and one-hot category lists
training_results.json	JSON	Metrics, accuracy, graphs, training history

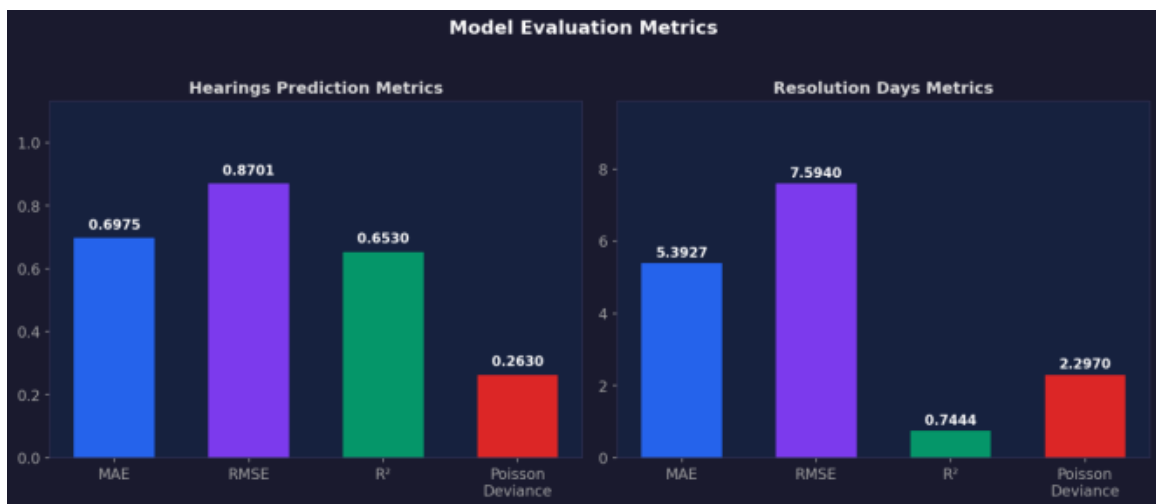
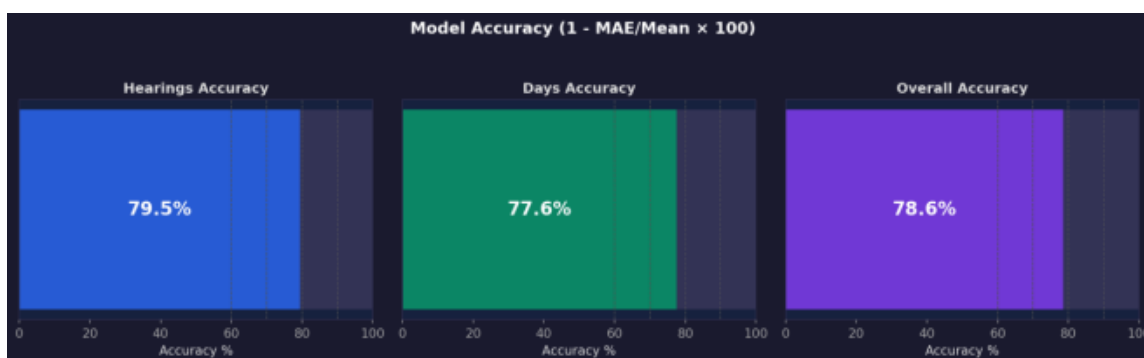


Figure 6: Model Evaluation Metrics

Metric	Hearings	Days	What It Means
MAE	0.7000	5.2910	Average abs error: ± 0.70 hearings, ± 5.29 days
RMSE	0.8693	7.4976	Root of squared error; some outliers exist
R ²	0.6536	0.7508	Explains 65.4% / 75.1% variance
Poisson Dev.	0.2639	2.2036	Count data fit; lower is better



The accuracy metric used is MAE-based relative accuracy:

$$\text{Accuracy} = (1 - \text{MAE} / \text{Mean}) \times 100$$

Step-by-Step Derivation

Hearings Accuracy:

1. Test set mean ≈ 3.40
2. MAE = 0.7000
3. MAE/Mean = 0.2059
4. Accuracy = $(1 - 0.2059) \times 100 = 79.41\%$

Days Accuracy:

1. Test set mean ≈ 24.09
2. MAE = 5.2910
3. MAE/Mean = 0.2196
4. Accuracy = $(1 - 0.2196) \times 100 = 78.04\%$

4.4.2 Prediction Pipeline

User Input $\rightarrow$ Validate $\rightarrow$ Encode $\rightarrow$ Scale $\rightarrow$ Predict $\rightarrow$ Clamp $\rightarrow$ Return

1. Input: Case_Type, Severity, Relationship, Num_Complainants, Num_Respondents, Lockdown_Status
2. Severity converted: String $\rightarrow$ numeric (Low=1, Medium=2, High=3)
3. Per-settlement prediction: Model predicts for each of the 4 settlement types
4. Feature encoding: OneHotEncoder + StandardScaler applied

5. Model inference: Forward pass produces raw predictions

6. Post-processing: Hearings clamped to 1–6, Days minimum 1, Weeks = days/7

4.4.3 Retraining & Fine-Tuning

Table 16: Two Training Modes

Mode	Trigger	Dataset	Process
Full Training	Manual (website)	CSV file (casedata.csv)	New model from scratch
Fine-Tuning	Auto/Manual	Django database	Loads existing, continues training

Table 17: Automatic Retraining Configuration

Setting	Default	Description
auto_retrain_enabled	True	Enable/disable auto retraining
threshold_cases	100	Resolved cases before trigger
default_epochs	50	Epochs for fine-tuning
default_validation_split	0.2	Validation split

Safety Mechanisms

- Versioned Backups: Every retrain creates a timestamped backup

- Automatic Rollback: If fine-tuning fails, previous model restored
- Minimum Data Check: Requires at least 50 resolved cases
- Cache Invalidation: Predictor cache cleared after retraining

4.4.3 Model Comparison

Four machine learning models were evaluated under identical conditions to determine the most suitable algorithm for the HearEase barangay mediation prediction system. Each model was trained on the same 959-sample training set, evaluated on the same 240-sample held-out test set, and measured using the same accuracy formula (MAE-based relative accuracy). No hyperparameter optimization was performed for any model to ensure a fair baseline comparison.

Model Overview

Table 18: Model Comparison

Model	Type	What It Does	How It Can Be Used for HearEase	Compatibility with HearEase
-------	------	--------------	---------------------------------	-----------------------------

Ridge Regression	Linear Model (L2 Regularized)	Finds the best linear combination of input features to predict each target. It assumes that the relationship between inputs (case type, severity, etc.) and outputs (hearings, days) is approximately linear. L2 regularization (penalty on large coefficients) prevents overfitting.	Could predict hearings and resolution days using a simple weighted sum of case features. Each feature gets a fixed coefficient (e.g., High severity adds +1.3 hearings). Predictions are fast (0.01s) and fully interpretable — barangay officials could see exactly why each prediction was made.	Low compatibility. Cannot learn incrementally from new resolved cases — requires full retraining from scratch. Cannot capture non-linear interactions (e.g., Family + High Severity having a disproportionate effect). HearEase's auto-retrain pipeline relies on incremental fine-tuning, which Ridge does not support.
Random Forest	Ensemble of Decision Trees (Bagging)	Builds 200 independent decision trees, each trained on a random subset of data. Each tree makes its own prediction, and the final output is the average of all trees. Naturally handles	Could predict case outcomes using ensemble voting from 200 decision trees. Provides built-in feature importance scores showing which case factors (severity, case type, relationship) are most influential.	Moderate compatibility. Handles non-linear patterns well, but cannot learn incrementally — every update requires retraining all 200 trees from scratch on the

		non-linear relationships and feature interactions without requiring special preprocessing.	Robust and hard to misconfigure.	full combined dataset. Trains two independent models (hearings and days separately) with no shared learning, which may produce logically inconsistent predictions (e.g., 1 hearing but 90 days).
XGBoost	Sequential Ensemble (Gradient Boosting)	Builds trees sequentially where each new tree corrects the errors of the previous ones. Includes built-in L1/L2 regularization. Widely considered the state-of-the-art for structured/tabular data in both industry and academic machine learning competitions.	Could provide highly accurate predictions on tabular case data. Offers SHAP-based interpretability for explaining individual predictions. Strong regularization prevents overfitting even on small datasets.	Moderate compatibility. Strong on static datasets but cannot learn incrementally — the entire sequential boosting chain must be rebuilt from scratch when new data arrives. Requires two completely separate models (hearings and days) with no shared representation. Hyperparameter management

				becomes more critical as the dataset evolves.
MLP (DNN) (This Study)	Multi-Output Neural Network	A deep neural network with a shared feature extraction trunk (128 → 64 neurons) and two independent prediction heads (one for hearings, one for days). Uses ReLU activation for non-linearity and dropout (20%) for regularization. Learns internal representations through backpropagation over the assigned training epochs.	Predicts both Total Hearings and Resolution Time Days simultaneously through a shared architecture. The shared trunk learns general case representations that benefit both predictions (multi-task learning). Can be incrementally fine-tuned with new resolved cases via the existing compare_and_retrain.py pipeline.	High compatibility. Native incremental fine-tuning allows the model to learn from newly resolved cases without full retraining. The shared-trunk architecture ensures hearings and days predictions remain logically consistent. Integrates directly with HearEase's auto-retrain trigger. Scalability improves with larger datasets.

Capability Comparison

The following table summarizes the technical capabilities of each model relevant to the HearEase system requirements:

Table 19: Comparison of Model Capabilities Against HearEase System Requirements

Capability	Ridge Regression	Random Forest	XGBoost	MLP (DNN) (This Study)
Non-linear pattern recognition	Cannot	Can	Can	Can
Feature interaction learning	Cannot	Can	Can	Can
Incremental fine-tuning	Cannot	Cannot	Cannot	Can
Multi-output joint learning	Cannot	Cannot	Cannot	Can
Interpretability	Full (coefficients)	Partial (feature importance)	Partial (SHAP values)	None (black box)
Training speed (1,199 samples)	0.01s (fastest)	0.24s (fast)	1.09s (moderate)	12.34s (slowest)
Overall accuracy	80.29%	79.06%	78.92%	78.92%

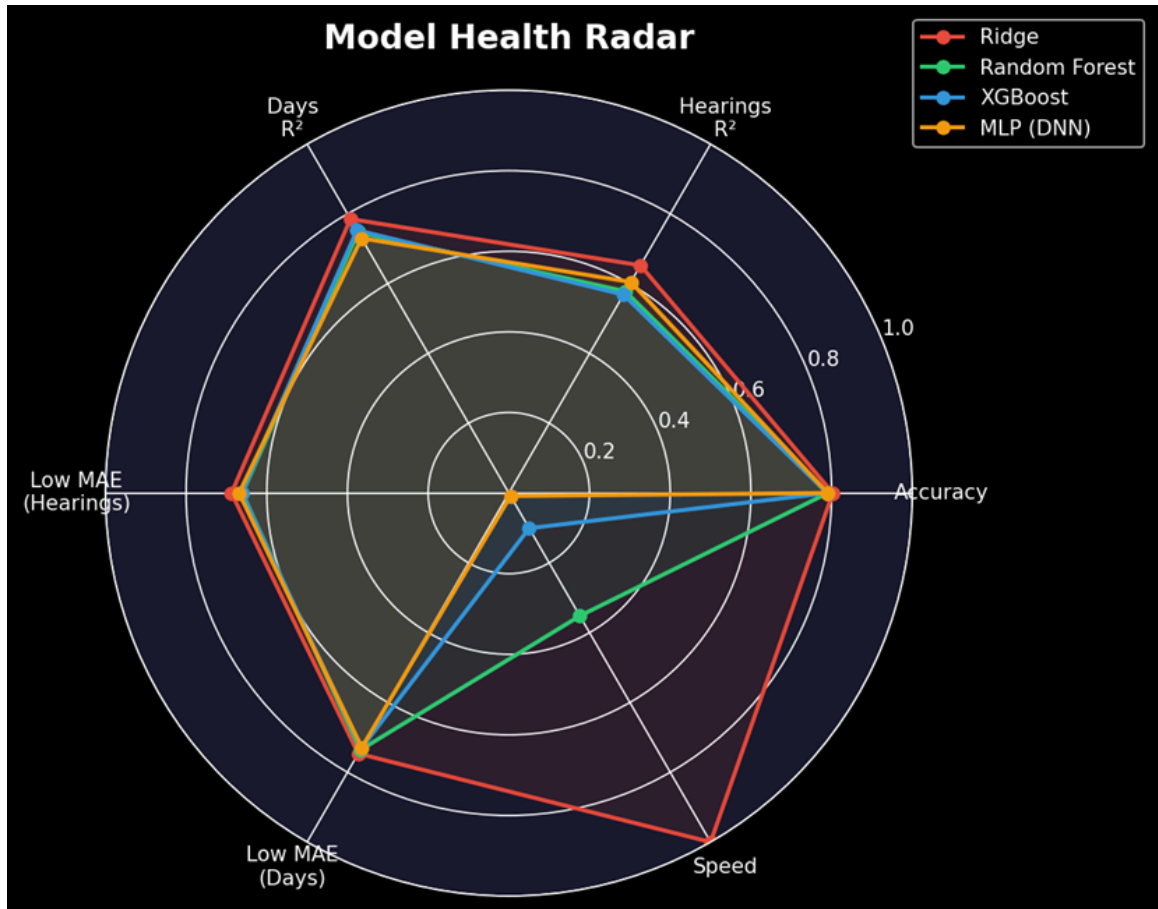


Figure 7: Radar Chart Comparison of Model Performance Metrics

Table 20: Performance Metrics Comparison of Evaluated Machine Learning Models

Model	Overall Accuracy	Hearings Accuracy	Days Accuracy	Hearings MAE	Days MAE	Hearings R^2	Days R^2	Time
Ridge	80.29%	81.62%	78.96%	0.625	5.0708	0.6524	0.7852	0.0092s

Rando m Forest	79.06%	80.02%	78.11%	0.6792	5.275	0.5779	0.7475	0.2376s
XGBoo st	78.92%	80.27%	77.57%	0.6708	5.4042	0.5703	0.7448	1.093s
MLP (DNN)	78.92%	79.78%	78.07%	0.6875	5.2833	0.5856	0.7546	12.3379 s

The Model Health Radar confirms that all four models perform comparably on static accuracy metrics, with Ridge Regression having the largest polygon area due to its speed and slight accuracy advantage. The maximum accuracy difference between the best model (Ridge, 80.29%) and worst model (XGBoost/MLP, 78.92%) is only 1.37 percentage points within the margin of statistical variation on a 240-sample test set.

The choice between these models cannot be made on accuracy alone. The MLP was selected for HearEase not because it is the most accurate on static data, but because it is the only model that supports all three operational requirements simultaneously, incremental fine-tuning with new resolved cases, shared multi-output learning that ensures logically consistent hearings-and-days predictions, and scalability to larger datasets where neural networks are expected to close the current 1.37% accuracy gap. Ridge Regression remains the strongest choice for any deployment where interpretability and training speed are prioritized over continuous learning.

4.4.3 Training Curves

The following graphs show model convergence over 50 epochs. Both training and validation curves are plotted to assess overfitting.

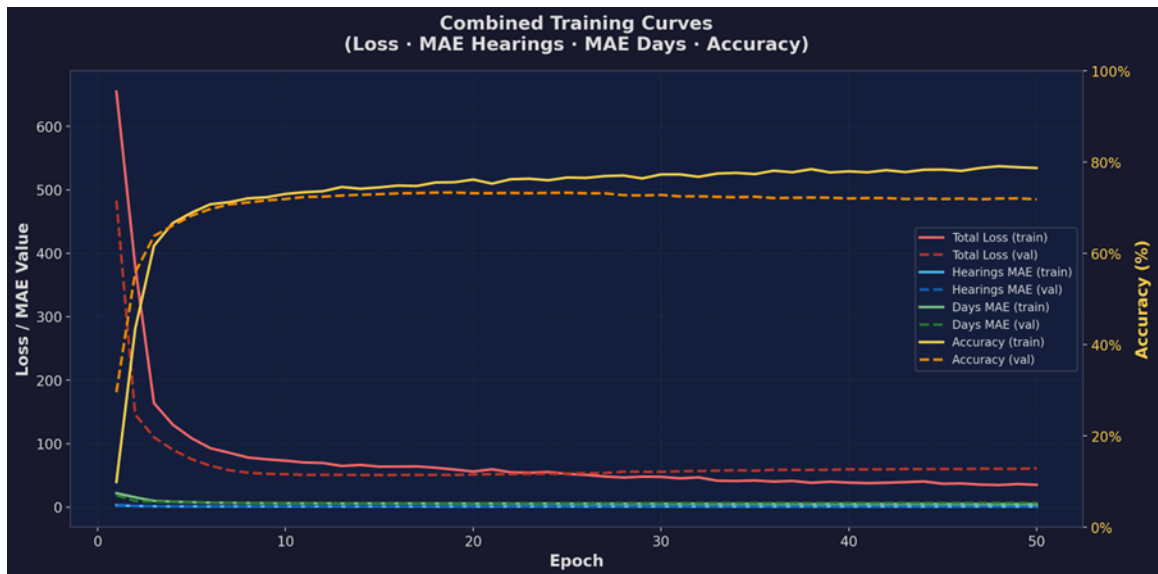


Figure 7: Model Convergence Analysis and Performance Metrics Across Training Epochs

The combined training curves graph confirms that the model trains effectively and generalizes well across all four tracked metrics:

- Total Loss drops sharply from 650 to 35 within the first 20 epochs and stabilizes without the validation loss ever increasing, no overfitting detected.

- Hearings MAE converges to 0.73(validation), predictions are within ± 1 hearing of the true value.
- Days MAE converges to 5.3(validation), predictions are within ± 5 days of actual resolution time. Given an average of 23 days per case.
- Accuracy rises from 30% to 78% by epoch 15 and plateaus. The small 4% gap between training (80%) and validation (76%) indicates a healthy generalization without memorization.

4.5 System Design and Implementation Features

4.5.1 The Centralized Administrative Dashboard

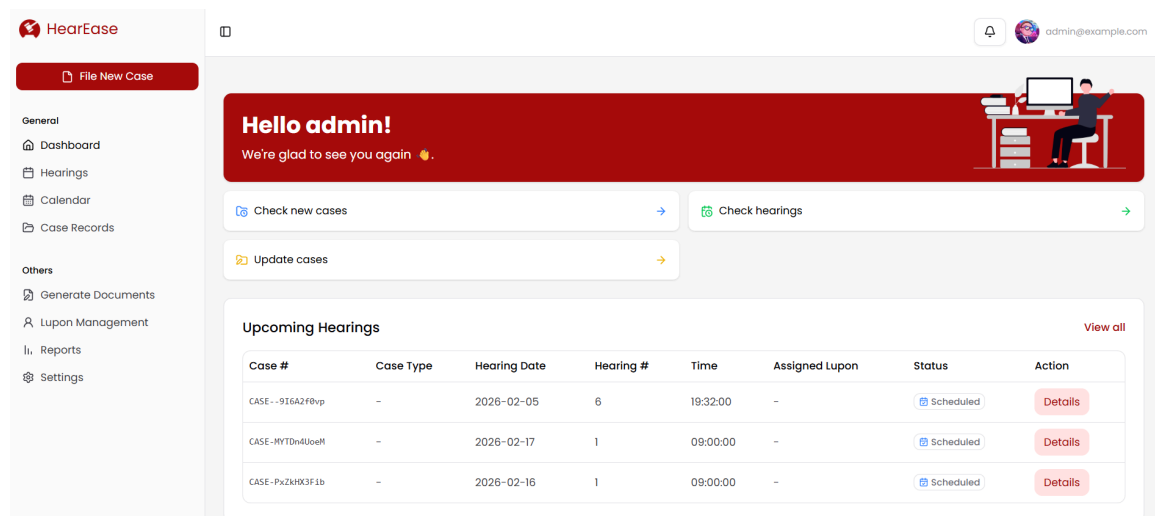


Figure 10: HearEase Administrative Dashboard

4.5.2 AI-Driven Case Filing and Prediction Interface

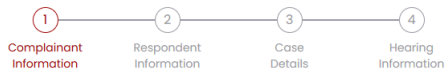


File New Case

Case Number: CASE-MYAdEhobIK

Date Filing: 2/16/2026

Start creating >



Complainant Information

+ Add Complainant

Q Find Complainant

Reset

Ruthiemy K Oribello

< Previous

Next >



Respondent Information

+ Add Respondent

Q Find Respondent

Reset

emman nunez

< Previous

Next >



Case Details

Nature of Complaint *

Select nature of complaint...

Severity Level *

Select severity level...

Relationship *

Select relationship...

Short Description *

Data Privacy Warning

You are responsible for ensuring that any uploaded media involving another person's face or personal information is taken with their consent. Unauthorized disclosure may violate the Data Privacy Act.

Upload Evidence

Upload images, videos, or documents as evidence. Videos up to 50MB allowed.

Click to upload file

Uploaded Files

No files uploaded.

< Previous

Next >



Hearing Information

Number of Predicted Hearing (auto) 3

Est. Resolution Time (auto) Approximately 2.3 weeks (16 days)

SCHEDULING INTERVAL

Standard

Expedite

Custom

1

Date 2/22/2026

Time 09:00 AM

Assigned Lupon Member Manuel Torres

2

Date 3/2/2026

Time 09:00 AM

Assigned Lupon Member Manuel Torres

3

Date 3/12/2026

Time 09:00 AM

Assigned Lupon Member Manuel Torres

< Previous

Submit Case >

Figure 11: AI-Driven Case Filing and Prediction Results

Figure 11 shows the sequential process of filing a new case. First, the system automatically retrieves the current date and generates a unique case number. Once confirmed, the system redirects to the Party Information Form, where the admin can add or search for Complainants and Respondents. In step 3, the admin chooses the nature of the complaint, changes the default severity value, selects the relationship status, provides a description, and optionally uploads case evidence. Lastly, in step 4, the AI prediction analysis is displayed. This shows the predicted number of hearings, estimated resolution time, and an editable schedule. It also recommends a Lupon member based on workload and shows available time slots for that day.

4.5.3 Intelligent Resource Allocation and Scheduling Recommendation

1 Complainant Information 2 Respondent Information 3 Case Details 4 Hearing Information

Hearing Information

Number of Predicted Hearing (auto) 3 Est. Resolution Time (auto) Approximately 2.3 weeks (16 days)

SCHEDULING INTERVAL Revert Changes to Default

☒ Standard Every 7 Days ☐ Expedite Less than 7 Days ☐ Custom Custom Interval

1 Date 2/23/2026 Time 09:00 AM Assigned Lupon Member (same for all hearings) Manuel Torres light

2 Date 3/2/2026 Time 09:00 AM Assigned Lupon Member (same for all hearings) Manuel Torres light

3 Date 3/9/2026 Time 09:00 AM Assigned Lupon Member (same for all hearings) Manuel Torres light

< Previous Submit Case >

Figure 12: Lupon Member Recommendation and Date Selection

Figure 12 presents the AI-driven prediction analysis. This shows the predicted number of hearings, estimated resolution time, and an editable schedule. It also recommends a Lupon member based on workload and shows available time slots for that day.

4.5.4 Digital Document Synthesis Engine

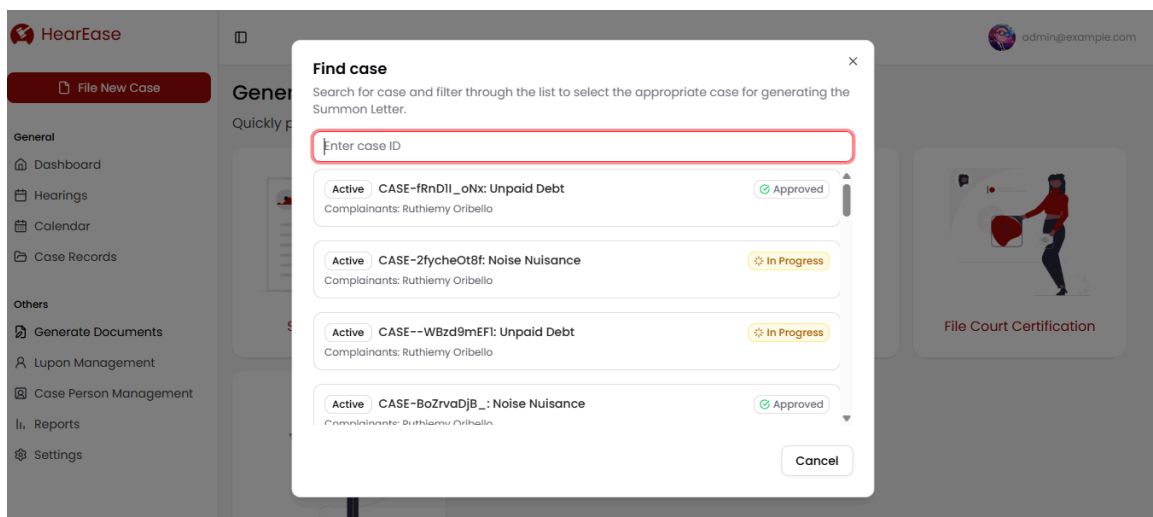
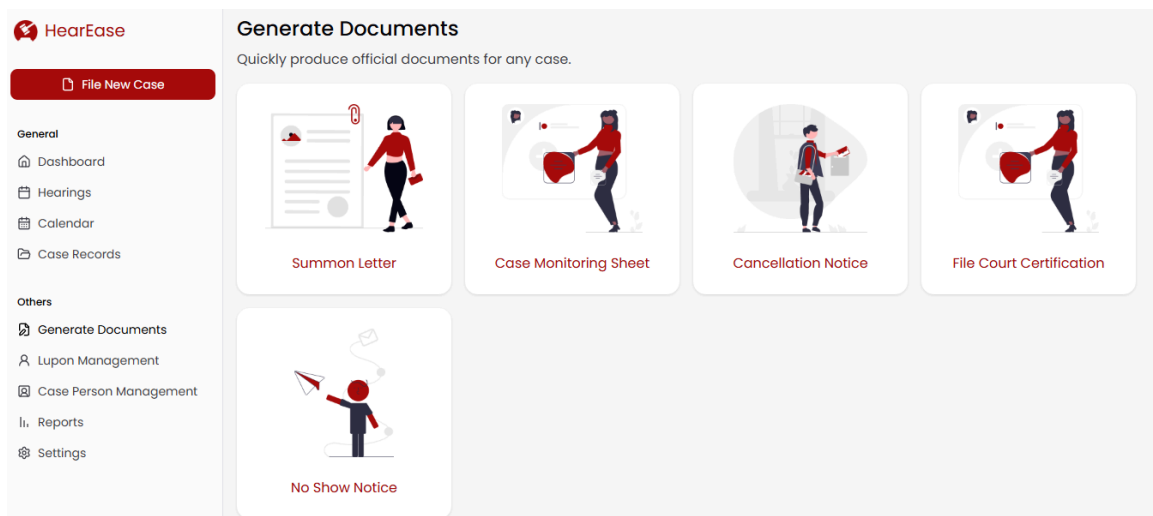


Figure 13: Automated Document Generation

Figure 13 displays the automated document generation feature. The system includes a default list of essential templates such as the Summon Letter, Case Monitoring Sheet, Notice of Cancellation, Certificate to File Action, and Notice of Non-Appeal. To generate a document, the administrator is prompted to select or search for the specific case, ensuring that the output is accurately linked to the correct file.

4.5.6 Public-Facing Resident Portal

Figure 14: Resident Case Filing and Tracking Portal

4.6 Assessment of Usability and Efficiency

The terminal phase of this study involved a summative evaluation of HearEase to determine its operational efficacy and usability relative to the legacy manual procedures in Barangay Tetuan. The researchers utilized a quantitative approach based on the ISO/IEC 25010 System and Software Quality Model, specifically targeting the dimensions of Functional Suitability and Usability.

To statistically measure the system's performance, a 5-point Likert scale was utilized. The following scale defines the mean ranges and their corresponding verbal interpretations used throughout the evaluation.

Evaluation Criteria	1 (Much Worse)	2 (Worse)	3 (Neutral)	4 (Better)	5 (Much Better)
1. Speed: How fast is it to secure a hearing schedule using the system compared to manual walk-ins?					
2. Ease of Use: How simple is the system's interface to navigate compared to traditional paperwork?					
3. Information Clarity: How clear is the AI-predicted resolution time compared to manual estimates?					

4. Effort Required: To what extent does this system reduce the "hassle" of filing and tracking a case?					
5. Overall Efficiency: How would you rate the system's ability to improve justice delivery in Barangay Tetuan?					

4.5.1 Expert User Evaluation Results

The Lupon Secretary evaluated the system's administrative capabilities, focusing on how the AI-driven backend and scheduling engine improved the clerical workflow.

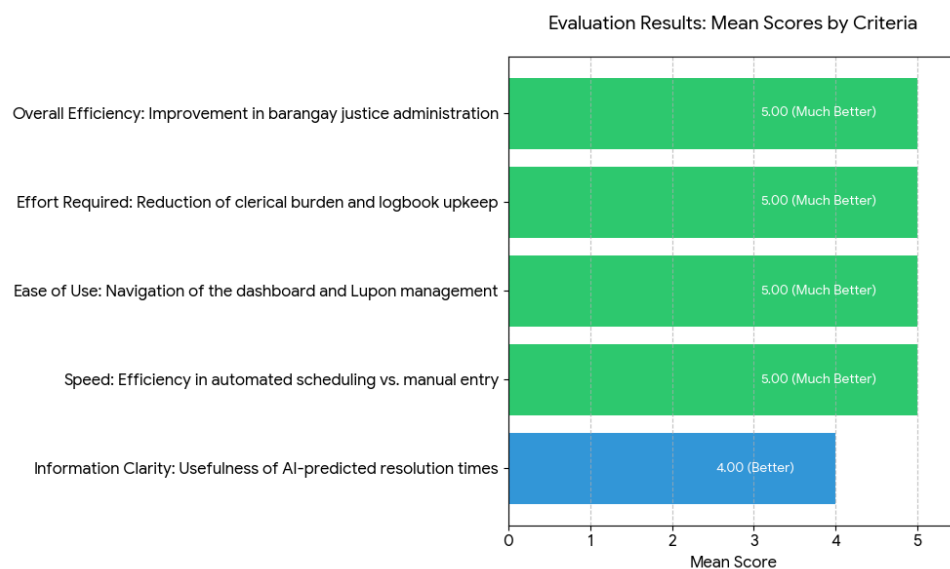


Figure 15: Expert User Evaluation Result

4.5.2 End-User Evaluation Results

A group of 25 citizen respondents evaluated the system. Since the AI logic is hidden from the public portal, the evaluation focused on the User Interface (UI), accessibility, and the transparency of case status updates.

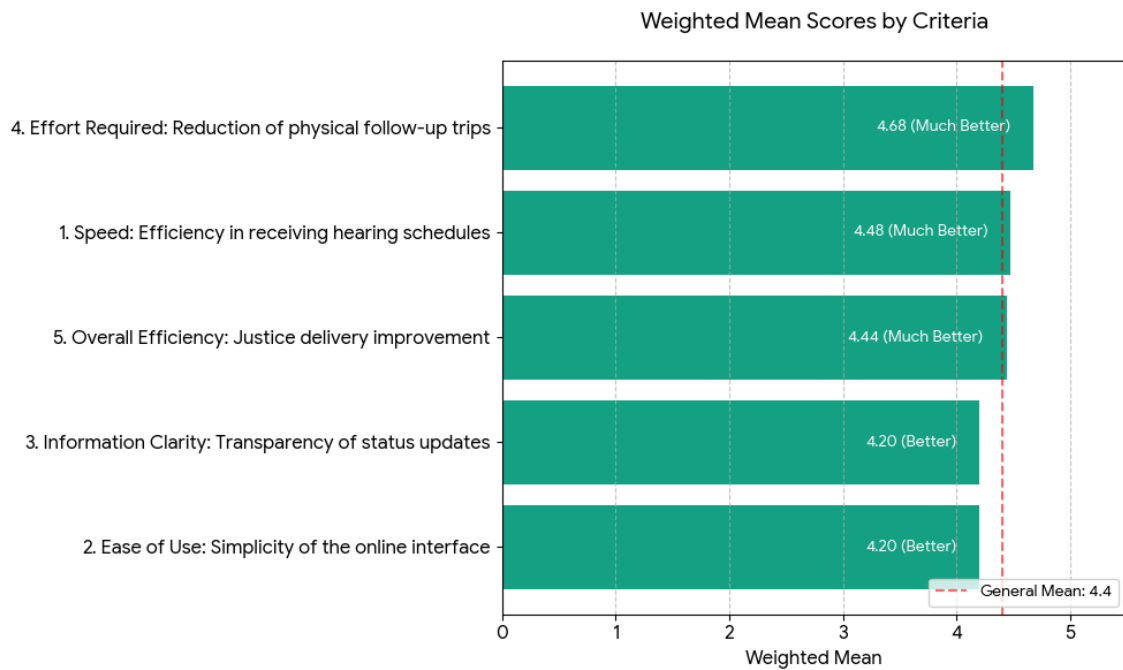


Figure 16: End User Evaluation Results

Table 16. Frequency Distribution of Citizen Evaluation Results

Evaluation Criteria	5 (Much Better)	4 (Better)	3 (Neutral)	2 (Worst)	1 (Much Worst)	Weighted Mean
1. Speed: How fast is it to secure a hearing schedule using the system compared to manual walk-ins?	16	6	2	1	0	4.48
2. Ease of Use: How simple is the system's interface to navigate compared to traditional paperwork?	11	9	4	1	0	4.20
3. Information Clarity: How clear is the AI-predicted	10	11	3	1	0	4.20

resolution time compared to manual estimates?						
4. Effort Required: To what extent does this system reduce the "hassle" of filing and tracking a case?	19	4	2	0	0	4.68
5. Overall Efficiency: How would you rate the system's ability to improve justice delivery in Barangay Tetuan?	14	8	3	0	0	4.44
General Weighted Mean						4.40

As shown in Table 16, while the system received a General Weighted Mean of 4.40 (Much Better), the distribution reveals specific areas where traditional methods are still valued by some segments of the community.

- **Effort Required (WM = 4.68):** This remains the highest-rated category. Even for respondents who were "Neutral" about the technology, they strongly agreed that the reduction in physical travel to the Barangay Hall for status updates was a significant benefit.
- **Ease of Use and Information Clarity (WM = 4.20):** These categories received several "Neutral" and one "Worse" rating. Qualitative interviews revealed that a small number of older residents found the digital forms and navigation of the entire system slightly intimidating compared to verbal reporting. Furthermore, one respondent noted that they preferred the traditional physical summons because they "trusted the paper more than a text message."
- **Speed (WM = 4.48):** The majority of users (88%) found the speed of scheduling. However, the few lower scores were attributed to intermittent mobile signal issues in certain parts of the barangay, which occasionally delayed the receipt of the SMS Gateway alerts.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

Summary of Findings

This study successfully developed and implemented HearEase, an AI-driven system utilizing Deep Neural Networks (DNN) to optimize barangay hearing scheduling and predict case resolution times in Barangay Tetuan, Zamboanga City. The research addressed critical inefficiencies in traditional manual scheduling processes that often resulted in delays, scheduling conflicts, and administrative burden on barangay officials and the Lupong Tagapamayapa.

Key findings from the study include:

System Development and Functionality

HearEase was successfully designed as a web-based platform with role-based access for barangay officials and a transparent case-tracking portal for residents. The system automates hearing scheduling based on established barangay rules, predicts case resolution times, and provides data-driven insights for operational optimization.

Predictive Accuracy

The Deep Neural Network model demonstrated strong predictive capabilities in forecasting case resolution times and the number of hearings required. Through comparative analysis of multiple neural network architectures, the selected DNN model achieved optimal performance in terms of accuracy, processing speed, and reliability when trained on historical case data from Barangay Tetuan.

User Acceptance and Satisfaction

User Acceptance Testing (UAT) revealed positive feedback from both barangay officials and the Lupong Tagapamayapa. Participants reported that the system was user-friendly, intuitive, and significantly improved their workflow. The centralized calendar and notification features were particularly valued for keeping all stakeholders informed of upcoming hearings and deadlines.

Data-Driven Decision Making

The reporting and analytics features of HearEase enabled barangay officials to monitor case resolution timelines, identify procedural bottlenecks, and make evidence-based decisions to improve service delivery. Historical data analysis revealed patterns in case types, resolution durations, and common scheduling challenges.

Transparency and Accessibility

The public portal component of HearEase enhanced transparency by allowing residents to file complaints digitally and track their case status in real-time. This feature improved citizen engagement and trust in the barangay justice system while maintaining compliance with legal requirements for in-person document service and hearings.

Scalability Potential

The pilot implementation in Barangay Tetuan demonstrated that the system architecture and methodology could be adapted and scaled to other barangays with similar governance structures and caseloads, contributing to broader improvements in local justice administration across the region.

Conclusions

Based on the findings of this research, the following conclusions are drawn:

AI Integration Enhances Barangay Justice Administration

The successful implementation of HearEase demonstrates that artificial intelligence and Deep Neural Networks can be effectively integrated into grassroots-level governance systems in the Philippines. The technology addresses longstanding inefficiencies in manual processes while respecting

established legal frameworks and cultural practices inherent in the Katarungang Pambarangay system.

Predictive Analytics Improves Resource Allocation

The ability to predict case resolution times and hearing requirements enables barangay officials to allocate resources more effectively, plan schedules proactively, and manage caseloads with greater precision. This predictive capability transforms reactive case management into a proactive, data-informed process.

Technology Adoption Requires User-Centric Design

The positive user acceptance results underscore the importance of designing systems that prioritize usability and align with the actual workflows of barangay officials. Success in technology adoption at the local government level depends not only on technical sophistication but also on intuitive interfaces and adequate training support.

Digital Transformation Enhances Transparency and Trust

By providing residents with direct access to case information and status updates, HearEase strengthens transparency in the barangay justice system. This transparency builds community trust and encourages greater civic engagement, contributing to more responsive and accountable local governance.

Historical Data is Valuable for System Training

The quality and comprehensiveness of historical case records directly impact the accuracy of AI predictions. Barangays that maintain detailed, well-organized records are better positioned to benefit from AI-driven systems, highlighting the importance of consistent documentation practices.

Hybrid Approach Balances Innovation and Tradition

HearEase successfully demonstrates that technological innovation can complement rather than replace traditional dispute resolution practices. The system enhances administrative efficiency while preserving the human-centered, community-based nature of barangay hearings and the essential role of the Lupong Tagapamayapa in mediation and conciliation.

Pilot Implementation Validates Broader Applicability

The success of the pilot study in Barangay Tetuan provides strong evidence that similar AI-driven scheduling and prediction systems can be implemented in other barangays throughout Zamboanga City and potentially across the Philippines, contributing to nationwide improvements in local justice administration.

Recommendations

Based on the conclusions of this study, the following recommendations are proposed for various stakeholders:

For Barangay Tetuan:

- Continue using and refining HearEase, incorporating ongoing feedback from users to enhance system features and address emerging needs.
- Invest in regular training sessions for barangay staff and Lupong Tagapamayapa members to ensure sustained proficiency in system use and maximize its benefits.
- Maintain rigorous data quality standards in case documentation to ensure the continued accuracy of AI predictions and system reliability.
- Monitor system performance metrics regularly and conduct periodic evaluations to assess long-term impact on case resolution efficiency and community satisfaction.

For Other Barangays in Zamboanga City:

- Consider adopting HearEase or similar AI-driven scheduling systems, adapting the platform to local governance structures, caseload characteristics, and resource availability.

- Establish partnerships with academic institutions and technology providers to facilitate knowledge transfer and technical support during implementation.
- Prioritize the digitization of historical case records to build robust datasets that can support effective AI model training.
- Engage in inter-barangay knowledge sharing to learn from Barangay Tetuan's implementation experience and best practices.

For Local Government Units (LGUs) and the Department of the Interior and Local Government (DILG):

- Develop policy frameworks and guidelines to support the adoption of AI-driven systems in barangay justice administration across the Philippines.
- Allocate funding and resources for technology infrastructure, training programs, and ongoing technical support to enable widespread implementation.
- Establish standards for data management, privacy protection, and ethical AI use in local governance contexts.
- Create incentive programs to encourage barangays to modernize their administrative processes and adopt evidence-based technologies.
- Facilitate regional and national forums for sharing successful case studies and lessons learned from AI implementation in local governance.

For Future Researchers:

- Conduct longitudinal studies to assess the long-term impact of AI-driven scheduling systems on case resolution outcomes, community satisfaction, and barangay operational efficiency.
- Explore the integration of additional AI capabilities, such as natural language processing for automated document generation or sentiment analysis for assessing community feedback.
- Investigate the applicability of HearEase methodology to other aspects of barangay governance beyond hearing scheduling, such as project monitoring, resource allocation, or citizen service delivery.
- Examine the scalability of the system to barangays with different demographic profiles, caseload volumes, and technological readiness levels.
- Research the ethical implications of AI use in justice administration, including issues of algorithmic bias, data privacy, and the balance between automation and human judgment.
- Develop frameworks for evaluating the cost-effectiveness and return on investment of AI implementations in local government settings.

For Technology Developers and Academic Institutions:

- Continue refining Deep Neural Network architectures and training methodologies to improve prediction accuracy and system adaptability.

- Develop user-friendly interfaces and training materials that accommodate varying levels of technological literacy among barangay officials.
- Create modular, customizable system designs that can be easily adapted to the specific needs and contexts of different barangays.
- Establish partnerships with local government units to ensure that technology development is grounded in real-world operational needs and constraints.
- Contribute to the broader body of knowledge on AI applications in public administration through publications, conferences, and open-source initiatives.

For the Community and Residents:

- Actively engage with the HearEase public portal to file complaints, track case status, and provide feedback on system performance.
- Participate in community consultations and feedback sessions to help shape the ongoing development and refinement of the system.
- Advocate for transparency, accountability, and ethical use of technology in local governance.
- Support barangay officials in their efforts to modernize administrative processes and improve service delivery.

In conclusion, HearEase represents a significant advancement in the application of artificial intelligence to grassroots governance in the Philippines. By

optimizing barangay hearing scheduling and predicting case resolution times, the system enhances administrative efficiency, promotes transparency, and contributes to more timely and accessible justice for local communities. The success of this pilot implementation provides a strong foundation for broader adoption and continued innovation in AI-driven local governance solutions.

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Appendix A: Sample Data Collected

Date of Consultation: *February 28, 2025*

Location: Barangay Tetuan Hall, Zamboanga City, Philippines, 7000

TOPIC	DETAILS/TAKEAWAYS
Hearing Scheduling Process	A hearing schedule is assigned to each case, followed by a summon letter personally delivered by the Barangay Officials to the respondent's address. Delivering Summons in-person is mandatory, despite the use of any e-barangay systems.
Role of Lupon Members	The Lupon Member who receives the complaint is responsible for facilitating the hearing on its scheduled date.
Lupon Duty Schedule	If Lupon 1 is on duty on Monday, and receives a case, Lupon 1 is automatically obligated to handle the hearing next Monday, and on the next

	Monday if the hearing is not settled.
Types of Cases Handled	Minor disputes (e.g. property conflicts) are resolved through barangay mediation hearings.
Cases Immediately Forwarded to Police Authority	Severe offenses (e.g., murder and physical violence) are immediately escalated to the police or higher authorities, and not catered for barangay hearings.
Identified Challenges	Manual scheduling and tracking often leads to delays and inefficiency.
Suggested Improvements	The officials expressed interest in implementing an automated scheduling system to streamline processes and reduce manual work.

Appendix B: System Design

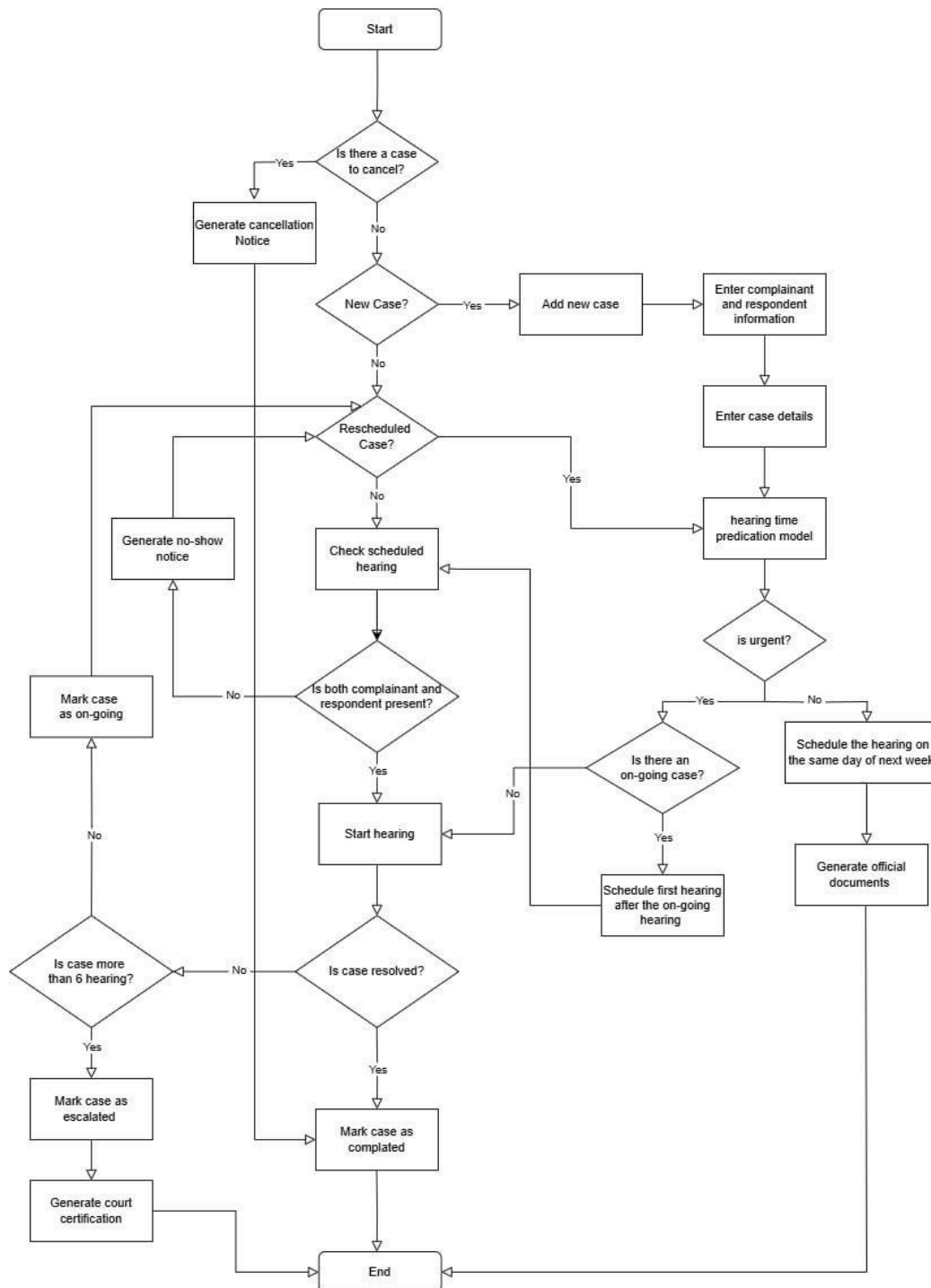


Figure 10: HearEase Flow Chart

Appendix C: Evaluation Tool

Instructions: Please rate your experience with the **HearEase System** compared to your past experiences with **Traditional Barangay Methods** by checking the box that best matches your opinion.

Evaluation Criteria	1 (Much Worse)	2 (Worse)	3 (Neutral)	4 (Better)	5 (Much Better)
1. Speed: How fast is it to secure a hearing schedule using the system compared to manual walk-ins?					
2. Ease of Use: How simple is the system's interface to navigate compared to traditional paperwork?					
3. Information Clarity: How clear is the					

AI-predicted resolution time compared to manual estimates?					
4. Effort Required: To what extent does this system reduce the "hassle" of filing and tracking a case?					
5. Overall Efficiency: How would you rate the system's ability to improve justice delivery in Barangay Tetuan?					

Appendix D: Relevant Source Code

```
def mlp(input_dim):
    # Shared Feature Extractor
    inp = Input(shape=(input_dim,), name="input_features")
    x = Dense(128, activation="relu", name="shared_dense_128")(inp)
    x = Dropout(0.2, name="shared_dropout")(x)
    x = Dense(64, activation="relu", name="shared_dense_64")(x)

    # Head for hearings
    h = Dense(32, activation="relu", name="hearings_dense_32")(x)
    out_hearings = Dense(1, activation="linear", name="hearings_output")(h)

    # Head for days
    d = Dense(32, activation="relu", name="days_dense_32")(x)
    out_days = Dense(1, activation="linear", name="days_output")(d)

    model = Model(inputs=inp, outputs=[out_hearings, out_days], name="global_case_model")
    model.compile(optimizer=Adam(), loss={"hearings_output": "mse", "days_output": "mse"},
                  metrics={"hearings_output": ["mae"], "days_output": ["mae"]})
    return model
```

Appendix E: Ethical Clearance

WMSU-REO-REC-001.2
Effective Date: 20-June-2025

 **WESTERN MINDANAO STATE UNIVERSITY**
Research and Extension
Research Ethics Office
accredited by
PHILIPPINE HEALTH RESEARCH ETHICS BOARD   

grants
this

RESEARCH ETHICS CLEARANCE

to

*Romelyn F. Dangaran
Emmanuel John Nuñez
Ruthiemy K. Oribello*

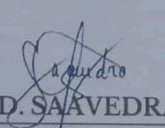
Research Title: HEAREASE: OPTIMIZING BARANGAY
HEARING SCHEDULING AND PREDICTING
CASE RESOLUTION TIME USING DEEP
NEURAL NETWORK

WMSU REO Code: 2025-UG-0834

The grant of this Research Ethics Clearance for Study Implementation mandates the researcher's adherence to the current WMSU REO processes and standards. It may be revoked anytime for any violations committed during the conduct of the study.

A Quarterly Progress Report must be submitted for continuing ethics review. This Research Ethics Clearance will expire on November 25, 2026.

Issued this 25th day of November 2025, at Research Ethics Office, Western Mindanao State University, Zamboanga City, Philippines.


ANALYN D. SAAVEDRA, Ph.D., J.D.
Research Ethics Director

Appendix F: Plagiarism Report

Appendix G: Research Critique and Editing Certificate

Appendix H: Curriculum Vitae

Ruthiemy K. Oribello

09455393908 | oribelloruthiemy@gmail.com | [linkedin.com/in/ruth-oribello](https://www.linkedin.com/in/ruth-oribello) | github.com/Rthuro | [Website Portfolio](#)

EDUCATION

Western Mindanao State University

Bachelor of Science in Computer Science

Normal Road, Baliwasan, Zamboanga City

Aug. 2022 – Present

EXPERIENCE

Lead UX/UI Designer and Frontend Developer

August 2024 – Present

Software Engineering Course

Western Mindanao State University, Z.C.

- Designed user-friendly and visually appealing interfaces using Figma
- Developed a responsive frontend using Svelte and Tailwind CSS for seamless adaptability across devices
- Implemented interactive UI components using shadcn for a modern and accessible user experience.
- Collaborated with a team of 6 to follow Waterfall methodologies and version control using GitHub.

Full-Stack Developer

November 2024 – March 2025

Advanced Database Systems Course

Western Mindanao State University, Z.C.

- Developed a fully functional hardware e-commerce system that operates locally using PHP, Tailwind CSS, SQL
- Designed an intuitive and responsive user interface with Tailwind CSS
- Integrated RESTful APIs for dynamic data handling and real-time updates.
- Tested and optimized system performance, ensuring smooth local operation.

Temporary Freelance Java Developer

December 2023

Object-Oriented Programming Course

Western Mindanao State University, Z.C.

- Developed local CRUD applications using Java Swing and SQL for Object-Oriented Programming coursework.
- Assisted two classmates by building functional Java applications for their academic requirements.
- Projects include:
 - * School Event Attendance System – A system to track and manage student attendance.
 - * McDonald's Employee Time Record – An application to log employee time-in and time-out.
 - * Simple Coffee Shop Ordering System – A basic POS system for order management.

PROJECTS

RePay - Lending Broker Platform | *Figma, Tailwind CSS, Svelte, shadcn*

August 2024 – Present

- Designed and developed the frontend using Svelte, Tailwind CSS, and shadcn for a modern and responsive UI
- Designed the website UI, including the admin, user, and lender interfaces, ensuring a seamless user experience
- Implemented reusable UI components for consistency and scalability across the platform

Azure Waves Café - Ordering System | *HTML, Tailwind CSS, JavaScript* | [LINK](#)

January 2025 - March 2025

- Built a frontend-only café ordering system using JavaScript with **local storage** to manage cart and user data.
- Developed a responsive menu and cart system where users can add, remove, and update items.
- Implemented a checkout process with receipt generation using **html2canvas**.
- Ensured smooth navigation with functional redirects and deployed on Netlify.

Hardware E-Commerce System | *PHP, Tailwind CSS, SQL, HTML*

November 2024 – March 2025

- Developed a fully functional e-commerce platform for hardware products using PHP, Tailwind CSS, SQL, and HTML in a local environment
- Implemented core e-commerce features including add-to-cart, checkout for pickup or delivery, and order history tracking
- Built an admin dashboard for managing products, orders, and users

CERTIFICATES

JavaScript Algorithms and Data Structures | *FreeCodeCamp* | [LINK](#)

March 2025

TECHNICAL SKILLS

Languages: PHP, SQL, JavaScript, HTML/CSS

Frameworks: Tailwind CSS, Bootstrap

Developer Tools: Git, VS Code, GitHub, phpMyAdmin

Design Tools: Figma, Canva

ROMELYN DANGARAN

PROJECT MANAGER | WRITER

Zamboanga City, Philippines, 7000 | romelyndangaran@gmail.com | 09694684985

SUMMARY

Highly organized and results-driven professional with a diverse background in administration, content creation, project coordination, and basic programming. Adept at effectively managing documents, leading small teams, and streamlining daily workflow. Passionate about maintaining organizational efficiency and leveraging user-friendly technology solutions to optimize work processes.

SKILLS

Workflow Optimization	Resource Allocation	Project Management Tools
Administration	Content Development	(Notion, Trello, Asana)
Decision-making	Digital content management	Basic Web Development (HTML,
Planning		CSS, PHP, SQL)

PROFESSIONAL EXPERIENCE

MANAGING EDITOR FOR ADMINISTRATION Jan 2024 - Present

- Oversee editorial processes, content management, and administrative functions
- Supervise and coordinate teams to ensure efficient workflow
- Develop and implement organizational strategies for publication operations
- Manage internal and external communications for streamlined productivity

PROJECT MANAGER - Resisync March 2024 - Present

- Resisync is a web-based application for apartemnt booking and management.
- Managed project timelines, deliverables, and resource allocation
- Coordinated between developers, designers, and stakeholders
- Oversaw system functionalities, ensuring smooth project execution
- www.resisync.site

CONTENT SPECIALIST - Queenship of Mary Prish January 2022 - Present

- Developed and curated content for parish publications and social media platforms
- Managed communication strategies to engage the parish community
- Assisted in administrative tasks related to content production and distribution

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE 2022 - 2026

Western Mindanao, State University, Zamboanga City

SCIENCE, TECHNOLOGY, ENGINEERING, MATHEMATICS 2019 - 2022

Southern City Colleges, Zamboanga City

TICTAPUL, NATIONAL HIGH SCHOOL 2019

Tictapul, Zamboanga City

ECHAGUE WEST CENTRAL SCHOOL 2013

Silauan Sur, Echague, Isabela

REFERENCES

Nephertilee Saddarani
Editor-in-Chief | Venom Publication
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Rev. Fr. Rogene Pervandos
Parish Priest | Queenship of Mary
Parish
<https://www.facebook.com/rogene.pervandos>

EMMANUEL JOHN NUÑEZ

09949864468 | nunezemanueljohn66@gmail.com | linkedin.com/in/emmanuel-john-nuñez-711ab3330 | <https://github.com/Lrioly>

EDUCATION

Western Mindanao State University

Normal Road, Baliwasan, Zamboanga City

Bachelor of Science in Computer Science

Aug. 2022 – Present

EXPERIENCES

Backend-Developer

Software Engineering Course

- Developed a robust and secure e-commerce website
- Collaborated with a team of 5 people as one of the back-end developers by following the Waterfall method and managing version control using GitHub
- Handled the admin side of the website
- Making a robust and efficient Data Flow Diagram for our Project

Advance Database Systeem Course

- Developed a website for a barangay in Zamboanga City for monitoring youth participation during events
- Collaborated with a front-end developer to create this website
- Optimized and tested the system, to ensure a smooth operation

Object-Oriented Programming Course

- Solely developed a robust and secure system to monitor and manage a user's accounts ensuring data privacy
- Utilized the Java programming language to make this system, including GUI and MySQL

PROJECTS

PConnect - A business to business e-commerce platform | Tailwind CSS, Laravel, etc.

- PConnect is a panel management system that manages panels from various distribution company in Zamboanga City, PConnect aims to enhance the efficiency of the booking process between store owners and distributors.

Safety Lock - Account manager | JAVA, JAVA GUI, MySQL

- Safety Lock is a system that protects any information that you put in it using a unique encryption method, which increases security and makes prevents data breaches and the data being stored in a remote server.

MNC - A simple e-commerce website for sellers and buyers | JavaScript, HTML, CSS, MySQL

- MNC is a simple e-commerce website for sellers and buyers. It is the smaller version of PConnect.

TECHNICAL SKILLS

Languages: PHP, SQL, JavaScript, C++, Python

Frameworks: Laravel, Bootstrap, Tailwind CSS

Developer Tools: Git, VS Code, Github, XAMPP

Design Tools: Figma, Canva